

Intelligent systems for industry, supply chain and environment IS4

LESSON 3

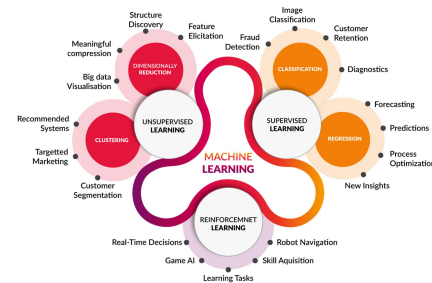
Narrow AI, Toolboxes, Taxonomy of AI systems and
Introduction to sensor in applications



Outline



- Prepare your toolboxes!
(Matlab, Colab, and the other resources)
- In this field, knowledge **interiorization** is needed to create **skills**
- Taxonomy of machine learning methods
- Focus on
 - Narrow AI
 - Regression
 - Classification
 - Clustering
 - Deductive and inductive learning
- Overview of AI applied to sensors
 - Intelligent sensors
 - Networks of sensors
 - Advanced instrumentation and **virtual sensors**

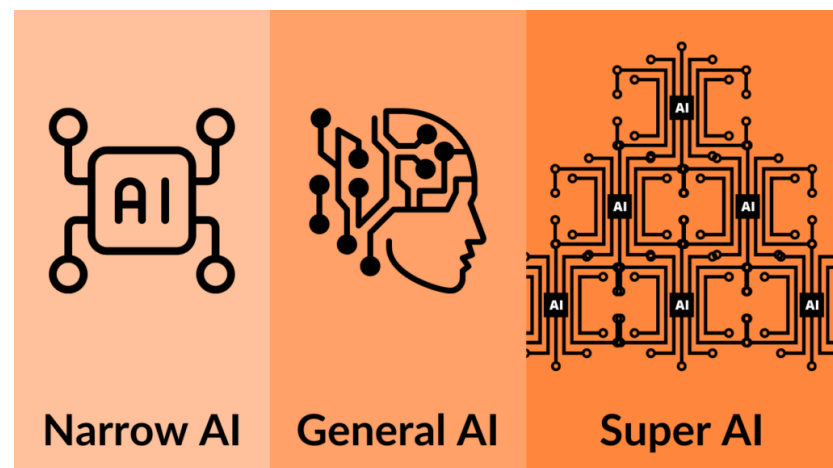




Narrow AI

General AI

Different points of view
which are present
in the state of the art



3 stages of AI



Narrow AI

Dedicated to assist with or take over specific tasks



General AI

Takes knowledge from one domain, transfers to other domain



Super AI

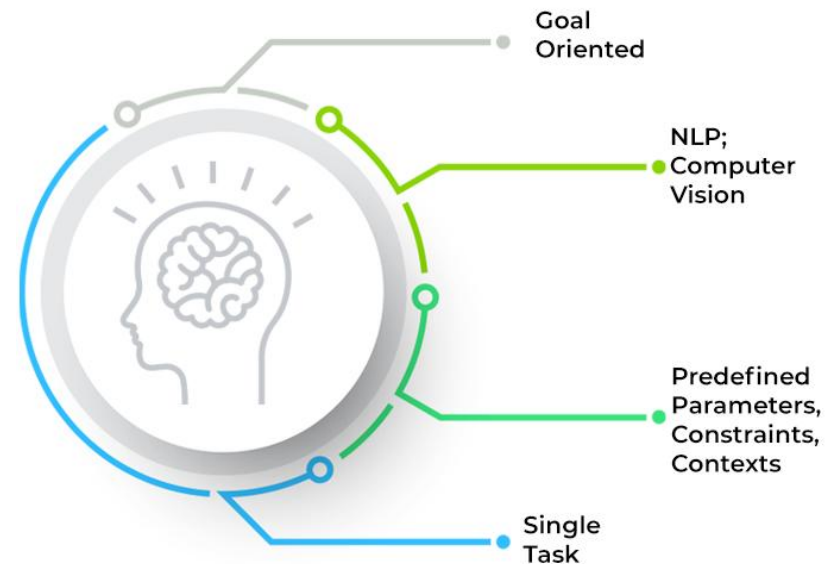
Machines that are an order of magnitude smarter than humans

Credit: Chris Noessel

Narrow AI

(or also Art. Narrow Int, → ANI)

- Also known as "Weak AI" refers to AI systems that are designed and trained to perform a specific task or set of tasks, typically within a narrow domain
 - For example, a machine learning model that is trained to recognize images of cats can only perform that specific task, and cannot generalize to other types of images or tasks
- Currently, Narrow AI is the predominant form of artificial intelligence in use today, and is applied to a wide range of applications such as speech recognition, image recognition, and natural language processing

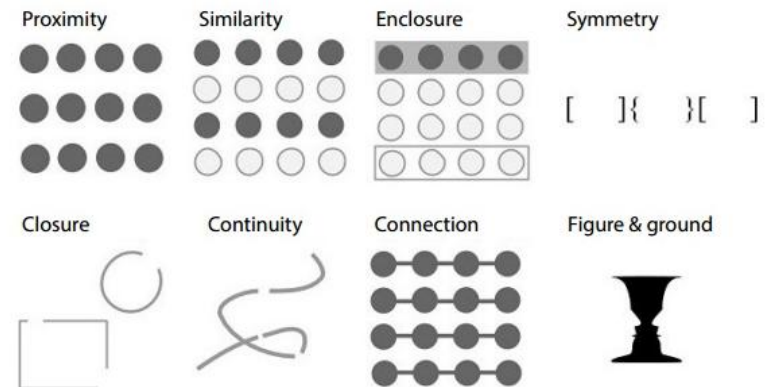


Typically, is MONO MODAL
(only images, only sound, etc..)

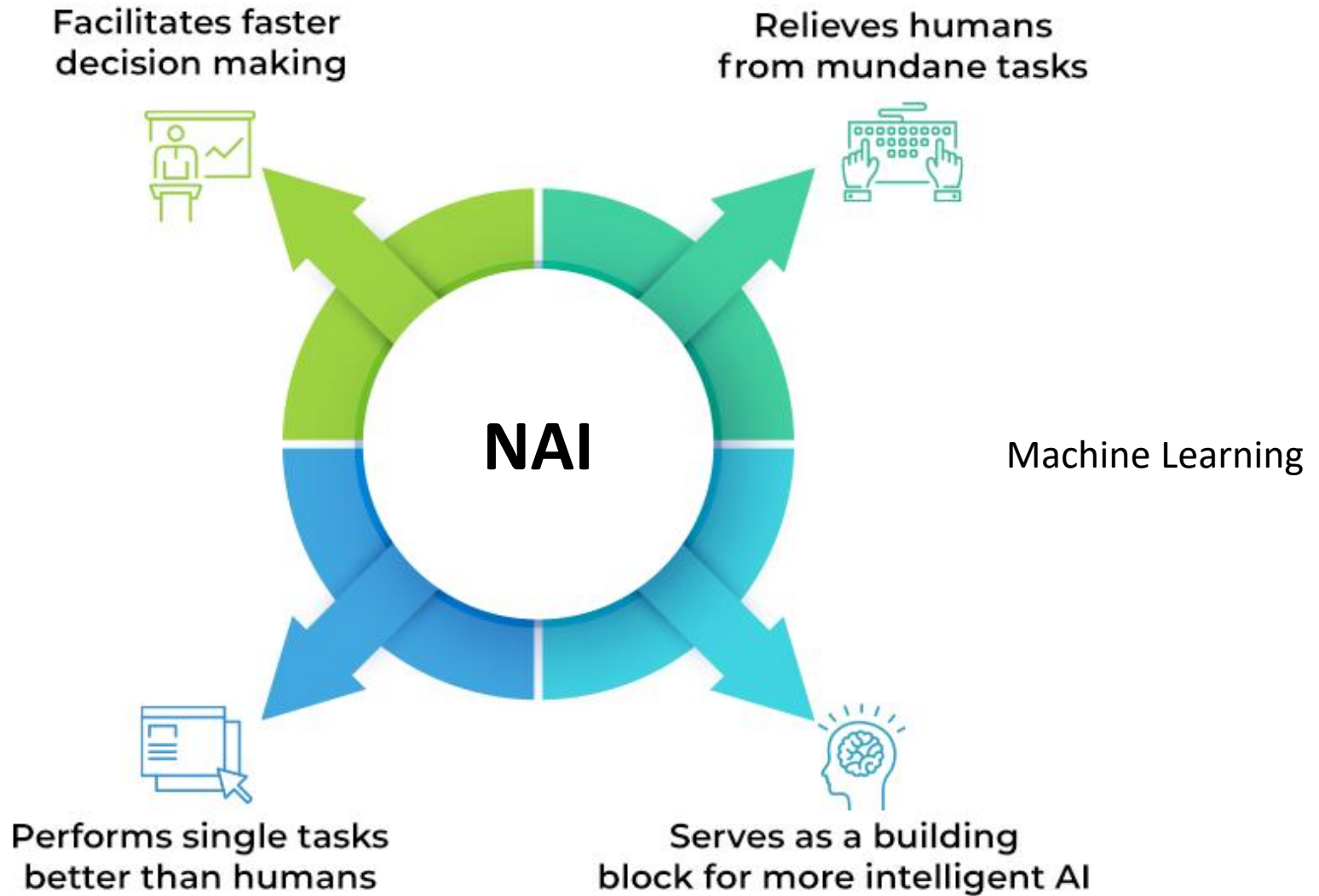
Narrow AI lacks...

- Self-awareness
- Consciousness
- Emotions
- Genuine intelligence that can match human intelligence
- and of course.... Gestalt

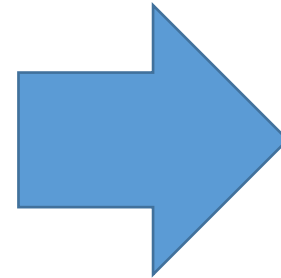
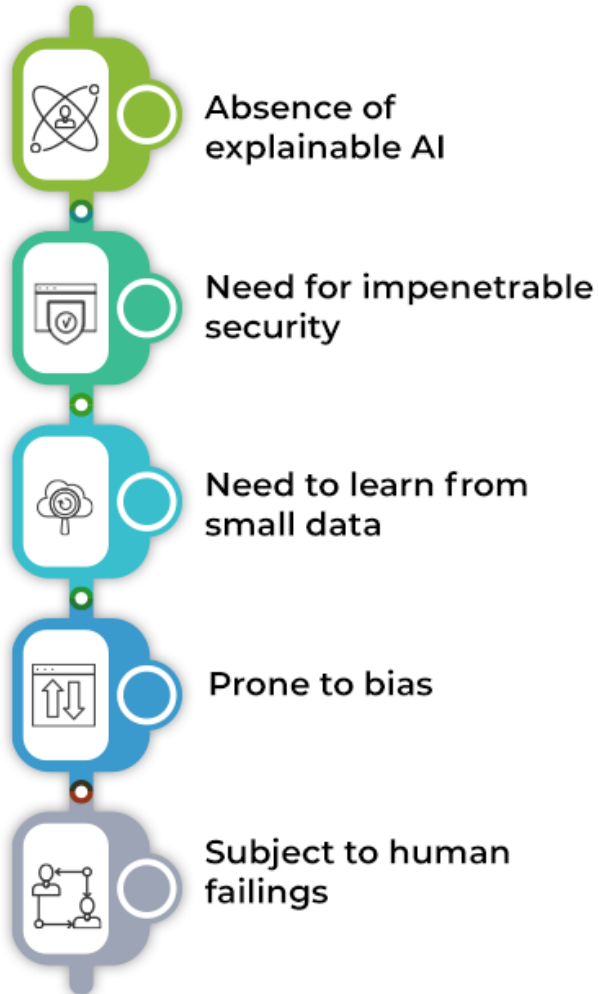
The capability to change dynamically the context, the level of details, the level of abstraction to better understand the input data according to your experience/knowledge



Advantages of Narrow AI



Challenges of Narrow AI



During this course we will review and test different techniques to start facing these challenges

General AI

(also known as Strong AI or Artificial General Intelligence → AGI)

- Refers to artificial intelligence systems that have human-like intelligence and the ability to reason, learn, and adapt to a wide range of tasks and situations, much like a human being.
- Would be capable of performing any intellectual task that a human can, and potentially even surpassing human capabilities in some areas.
- Is still a theoretical concept, by significant improvements are in progress.



Unorganized.... Let's start with your emails

<https://youtu.be/GV01B5kVsC0?t=207>

Since there is not a standard definition of what is «human-like» it will be hard to define the AGI

OS 1



Some statements about AGI...



- S. Altman, Jan 7, 2025
“My guess is we will hit AGI sooner than most people in the world think and it will matter much less”
- *“a few thousand days,”*
which translates to approximately 5–10 years
- *“in 2025, we may see the first AI agents ‘join the workforce’ and materially change the output of companies”.*
-*“we may see”* or *“it’s possible,”*
- The AI industry is highly competitive, with companies like OpenAI, Google (DeepMind), Meta, Anthropic, and xAI racing to achieve AGI

<https://time.com/7205596/sam-altman-superintelligence-agi/>

<https://blog.samaltman.com/reflections>

<https://www.pcmag.com/news/google-co-founder-says-60-hour-weeks-needed-to-achieve-agi>





SAM ALTMAN
CEO, OpenAI



JACOB ANDREAS
Professor, MIT



YOSHUA BENGIO
Professor, University of Montreal



RODNEY BROOKS
Professor emeritus, MIT, and cofounder, RobustAI



JOY BUOLAMWINI
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ALISON GOPNIK
Professor, UC Berkeley



GEOFFREY HINTON
Professor, University of Toronto



JARON LANIER
Computer scientist, entrepreneur, author, artist



GARY MARCUS
Professor, NYU



MELANIE MITCHELL
Professor, Santa Fe Institute



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CHRISTOF KOCH
Chief scientist of the MindScope program, Allen Institute



YANN LeCUN
Chief AI scientist, Meta



MARGARET MITCHELL
Chief ethics scientist, Hugging Face



ANDREW NG
Founder and CEO, Landing AI



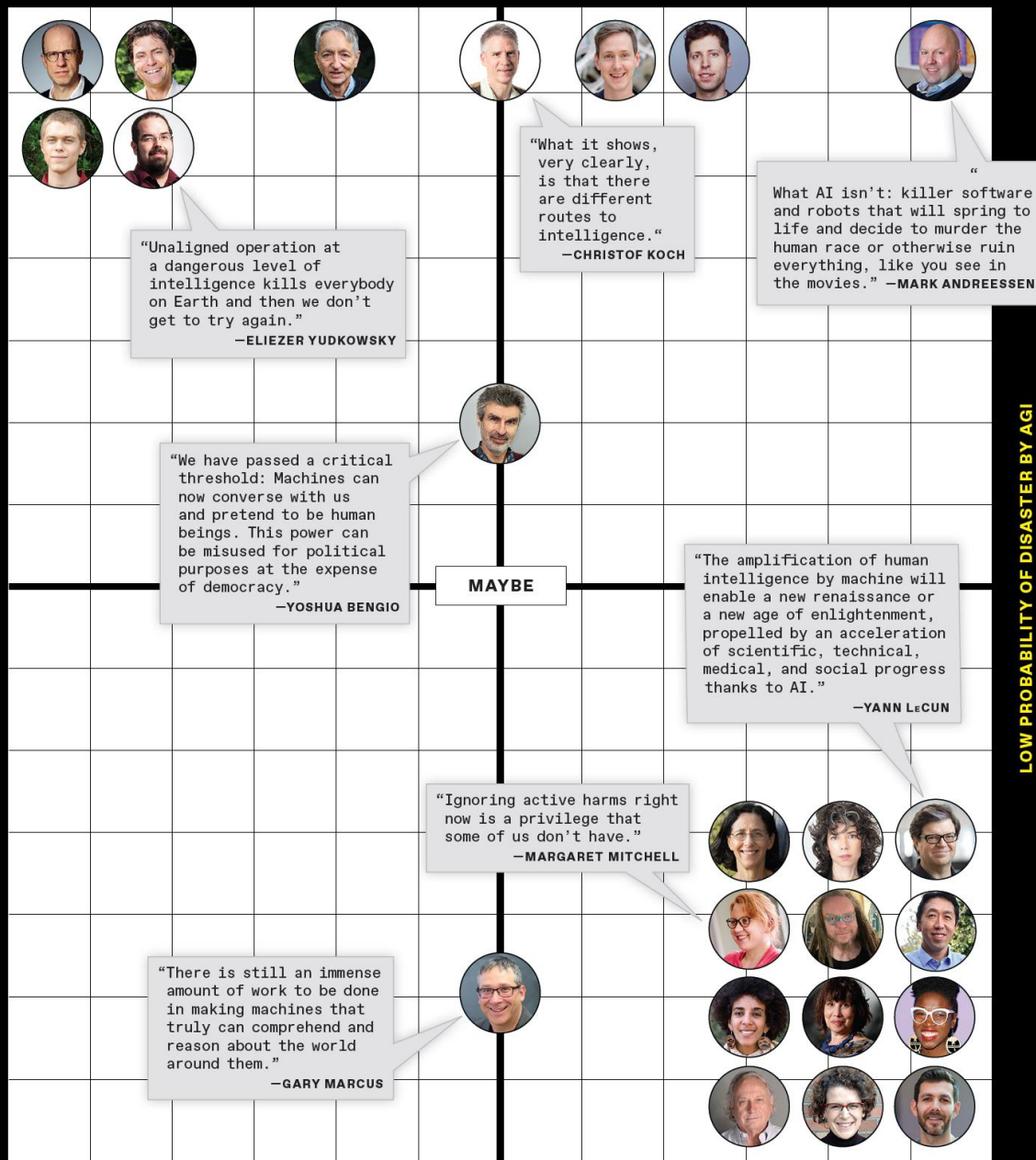
MEREDITH WHITTAKER
President, Signal Foundation

HIGH PROBABILITY OF AGI

Few more details about that...
the IEEE AGI panel

HIGH PROBABILITY OF DISASTER BY AGI

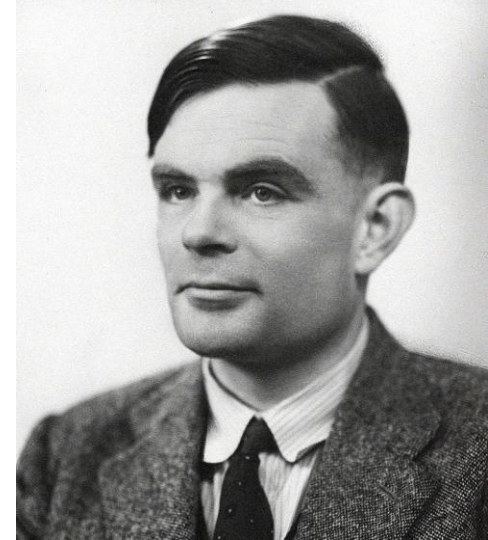
LOW PROBABILITY OF DISASTER BY AGI



LOW PROBABILITY OF AGI

AGI discussions

- **Main issues about AGI**
 - AGI is not a single agreed scientific threshold.
 - Different researchers mean different things by AGI
 - Human-level flexibility, transfer across domains, autonomous problem solving, or economic usefulness.
- **How AGI claims should be interpreted**
 - Scientific claim
 - Engineering capability claim
 - Economic impact claim
 - Public communication or strategic claim

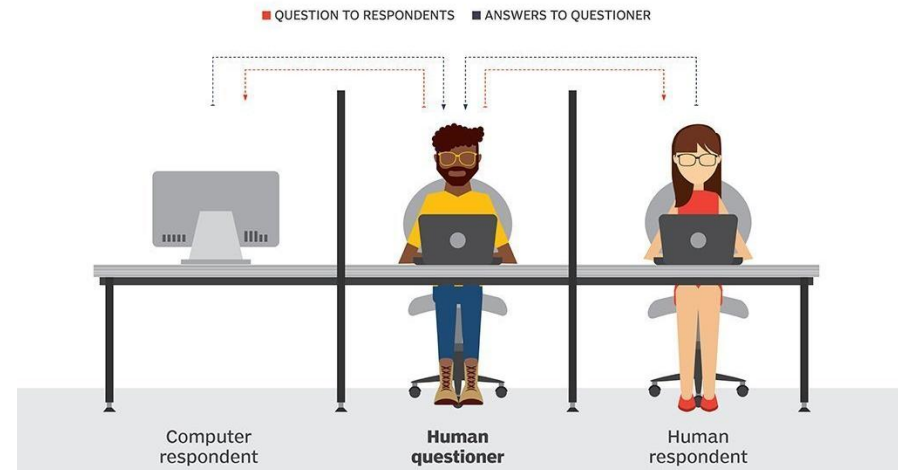


What about the TURING TEST?

Proposed by Alan Turing in 1950, is a measure of a machine's ability to exhibit intelligent behavior indistinguishable from that of a human.

- The test involves a human judge engaging in a natural language conversation with both a human and a machine without seeing them.
- If the judge cannot reliably tell the machine from the human, the machine is said to have passed the test, demonstrating a form of artificial intelligence.

During the Turing test, the human questioner asks a series of questions to both respondents. After the specified time, the questioner tries to decide which terminal is operated by the human respondent and which terminal is operated by the computer.



The Turing Test focuses on conversational ability, not general intelligence!

What about the TURING TEST?

Recent tests (Nature Journal) reports:

- Current AI systems can mimic human-like **conversation** convincingly, challenging traditional assessment methods
- ChatGPT has surpassed the Turing test, a benchmark for evaluating AI's human-like intelligence.
- This breakthrough has initiated a race among researchers to develop new methods for assessing AI capabilities.
 - There's a need for new evaluation methodologies that can better gauge AI's logic and reasoning skills
- Visual logic and reasoning tasks are highlighted as areas needing improved evaluation techniques for AI systems

<https://www.nature.com/articles/d41586-023-02361-7>

What about other tests?

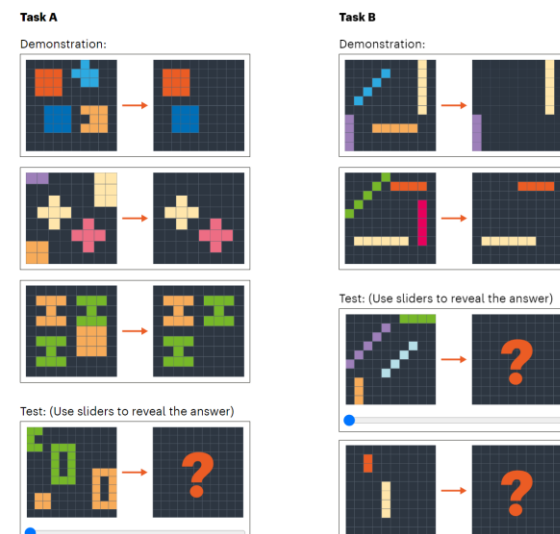
AI system performance — selected results

Test	GPT-4	GPT3.5	Specifically trained AI
Uniform Bar Exam	298/400 (~90th percentile*)	213/400 (~10th percentile)	N/A
Medical Knowledge Self-Assessment Program	75%	53%	N/A
Graduate Record Exam: Quantitative	163/170 (~80th percentile)	147/170 (~25th percentile)	N/A
HellaSwag: common-sense reasoning (test designed for machines)	95.3%	85.5%	85.6%
DROP: reading comprehension benchmark (for machines)	80.9%	64.1%	88.4%
GSM-8K: data set of 8,000 mathematics word problems at primary-school level (for machines)	92%	57.1%	87.3%

Source: OpenAI/Ref. 4.

*Percentile values are for human test-takers achieving this score.

An abstract-thinking test that defeats machines
Artificial-intelligence systems have so far been unable to achieve human-level performance on the ConceptARC test

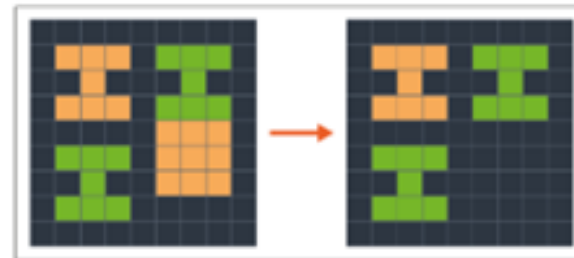
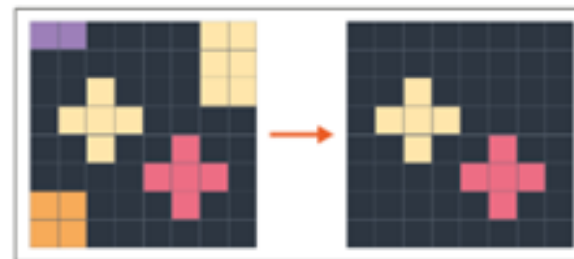
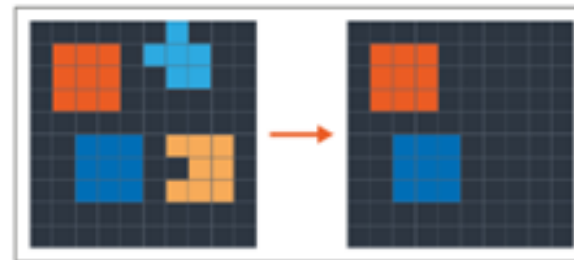


Wanna try?

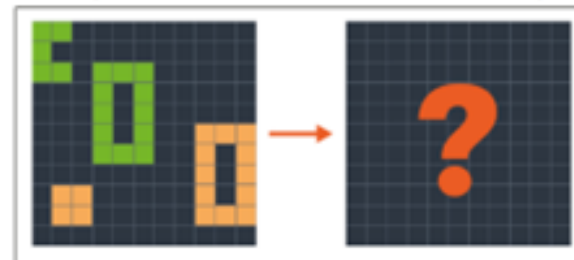
ConceptARC test

Task A

Demonstration:



Test: (Use sliders to reveal the answer)

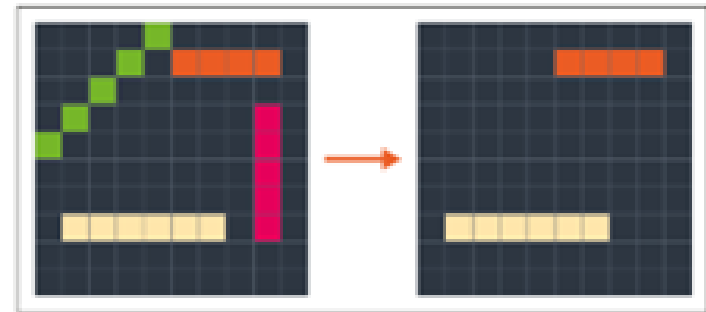
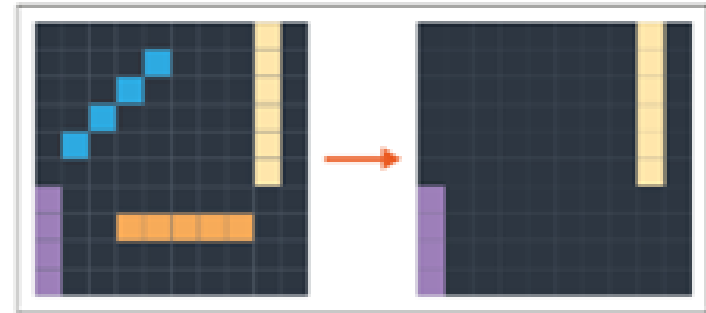


Wanna try?

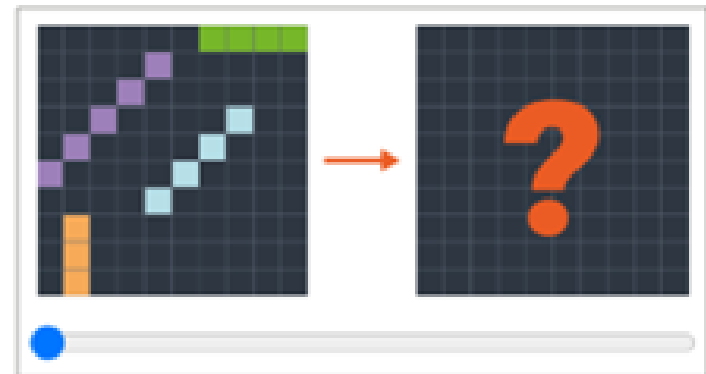
ConceptARC test

Task B

Demonstration:



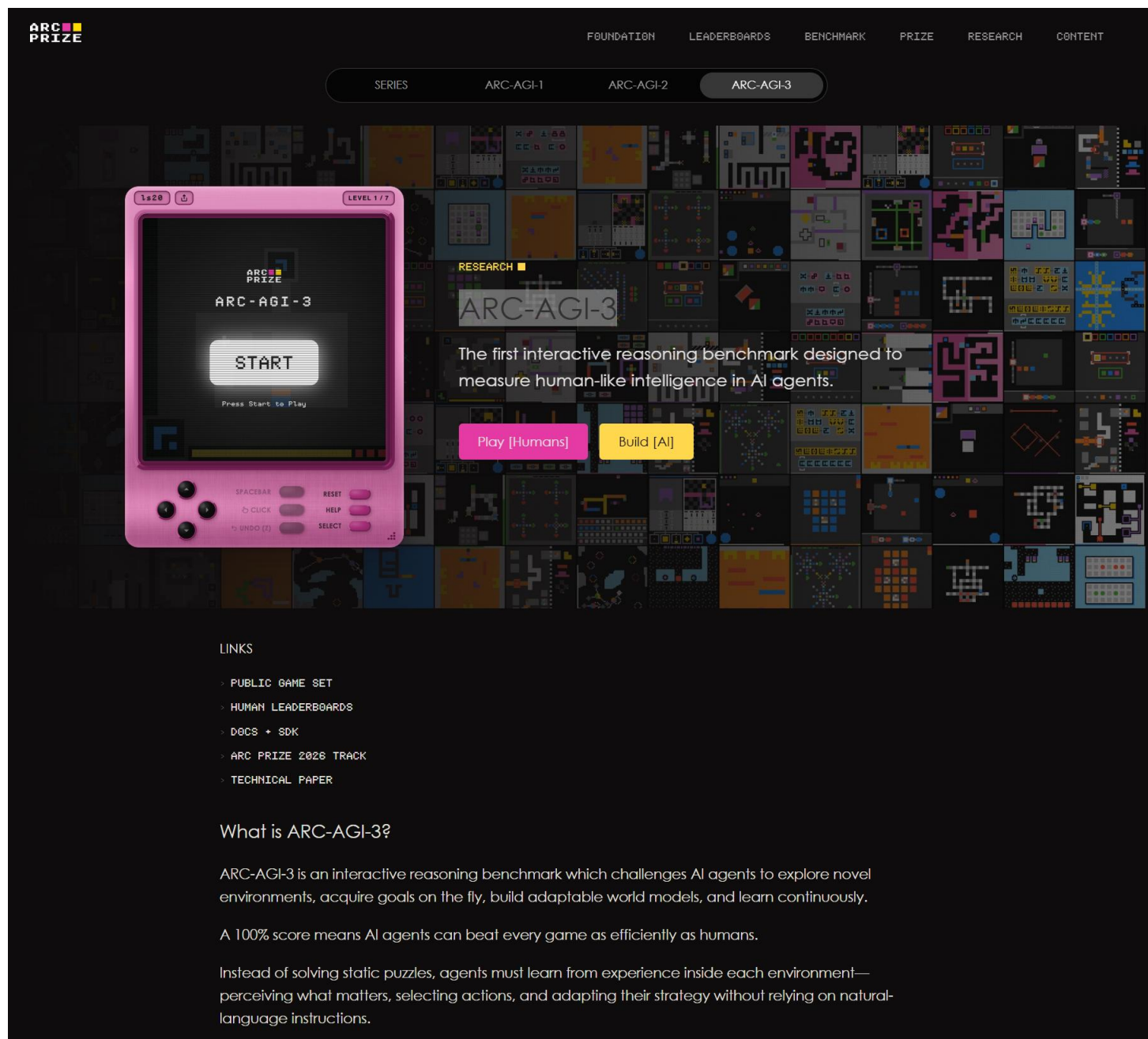
Test: (Use sliders to reveal the answer)



Ask yourself, what feature of my intelligence am I using to solve the puzzle?

- Random guess?
- Vision processing?
- Linguistic processing?
- Gestalt?

ARC AGI-3



ARC PRIZE

FOUNDATION LEADERBOARDS BENCHMARK PRIZE RESEARCH CONTENT

SERIES ARC-AGI-1 ARC-AGI-2 ARC-AGI-3

ARC PRIZE

ARC-AGI-3

START

Press Start to Play

RESEARCH

ARC-AGI-3

The first interactive reasoning benchmark designed to measure human-like intelligence in AI agents.

Play [Humans] Build [AI]

LINKS

- PUBLIC GAME SET
- HUMAN LEADERBOARDS
- DOCS + SDK
- ARC PRIZE 2026 TRACK
- TECHNICAL PAPER

What is ARC-AGI-3?

ARC-AGI-3 is an interactive reasoning benchmark which challenges AI agents to explore novel environments, acquire goals on the fly, build adaptable world models, and learn continuously.

A 100% score means AI agents can beat every game as efficiently as humans.

Instead of solving static puzzles, agents must learn from experience inside each environment—perceiving what matters, selecting actions, and adapting their strategy without relying on natural-language instructions.

Other recent tests

Benchmark	Input type	Metric	Frontier scores (2026)	What it measures
MMLU-Pro Massive Multitask Language Understanding — Pro Knowledge · Reasoning	Text only	Accuracy (%) 12k questions, 10 choices	GPT-5.4 ~86% Qwen 3.5 / Claude Opus 4.6 85–86% <i>Human expert: ~65%</i>	Graduate-level knowledge across 57 subjects (STEM, law, medicine, humanities). Upgraded from MMLU: 10-choice answers force reasoning, not guessing. Standard knowledge benchmark for frontier models.
MMMU-Pro Massive Multi-discipline Multimodal Understanding — Pro Vision + Text	Text + Image charts, diagrams, science figs	Accuracy (%) vision-only input settings	GPT-5.4 81.2% Gemma 4 31B 76.9% Claude Opus 4.6 ~76%	Multimodal reasoning across 30 academic disciplines. Filters out questions solvable by text alone — model must genuinely interpret the image. The reference benchmark for vision-language models.
GPQA Diamond Graduate-Level Google-Proof Q&A Expert Reasoning	Text only	Accuracy (%) PhD-level science	Qwen 3.5 / Gemini 2.5 Pro 83–85% Claude Opus 4.6 79.6% <i>Human expert: ~69% — models exceed humans</i>	Questions in physics, chemistry, biology designed to resist web search. "Google-proof" means the answer requires genuine reasoning, not retrieval. Frontier models now exceed the human expert baseline.
SWE-bench Verified Software Engineering Benchmark Coding Agentic	Text + Code GitHub issues, real repos	% resolved 500 verified issues	Claude Opus 4.6 80.8% GPT-5.4 ~80% <i>2023 baseline: 4.4% → 2026: 80%+</i>	Model resolves real GitHub issues end-to-end in real Python repos. Gold standard for agentic coding. One of the fastest progressions in AI benchmark history: from 4% to 80% in 3 years.

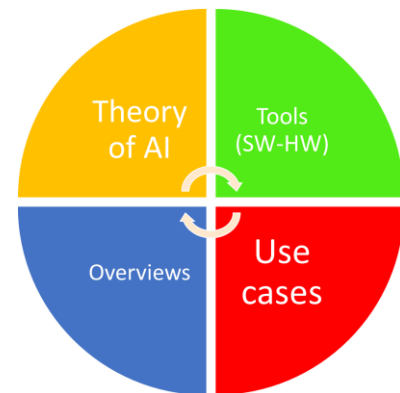
Sources: Anthropic model card, OpenAI API docs, Stanford AI Index 2025, BenchLM (April 2026).



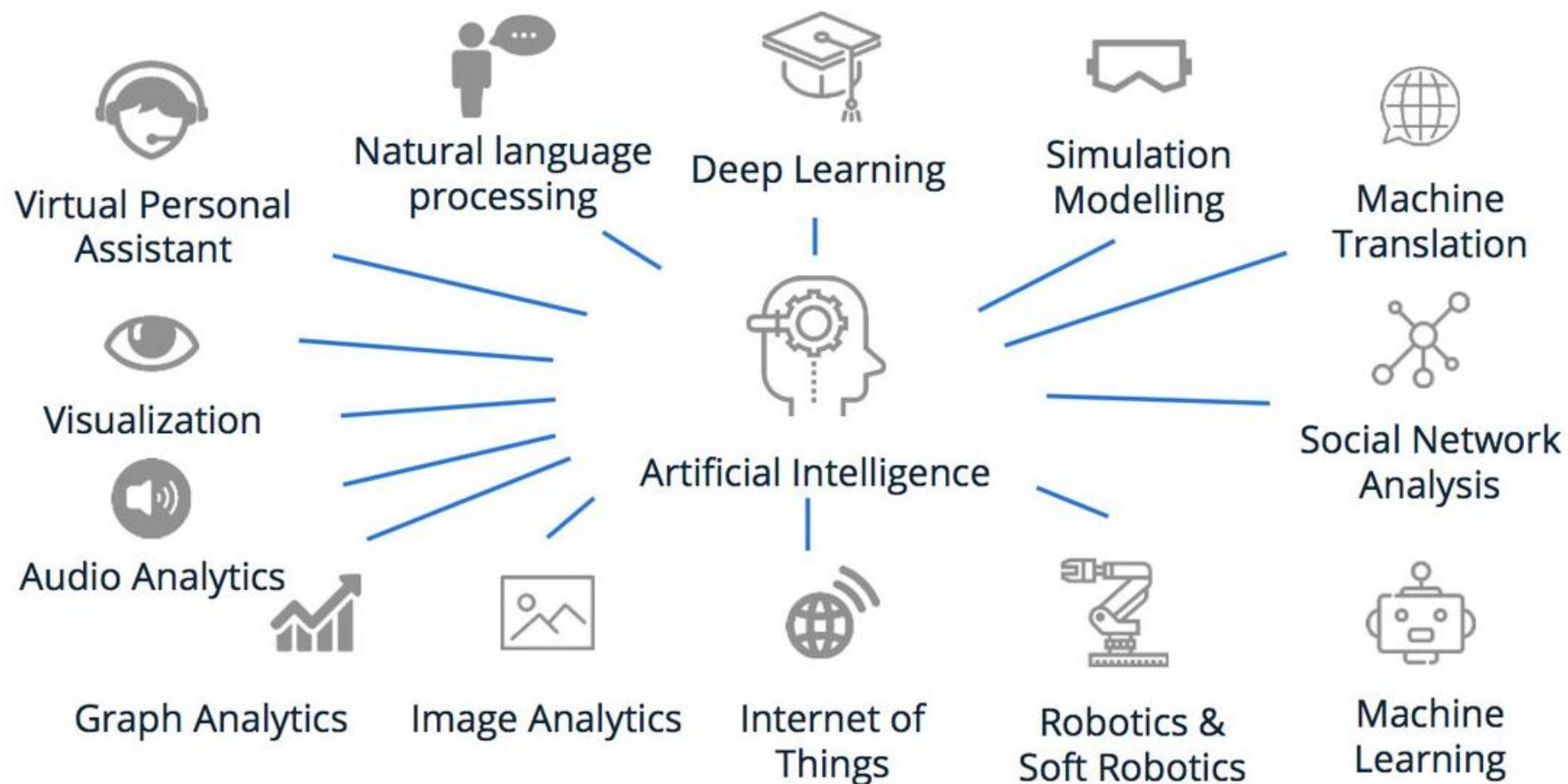
THEORY

Intelligent systems: main types

Many applications but under a general framework



Different topics ...



Mimiking what kind of worker?

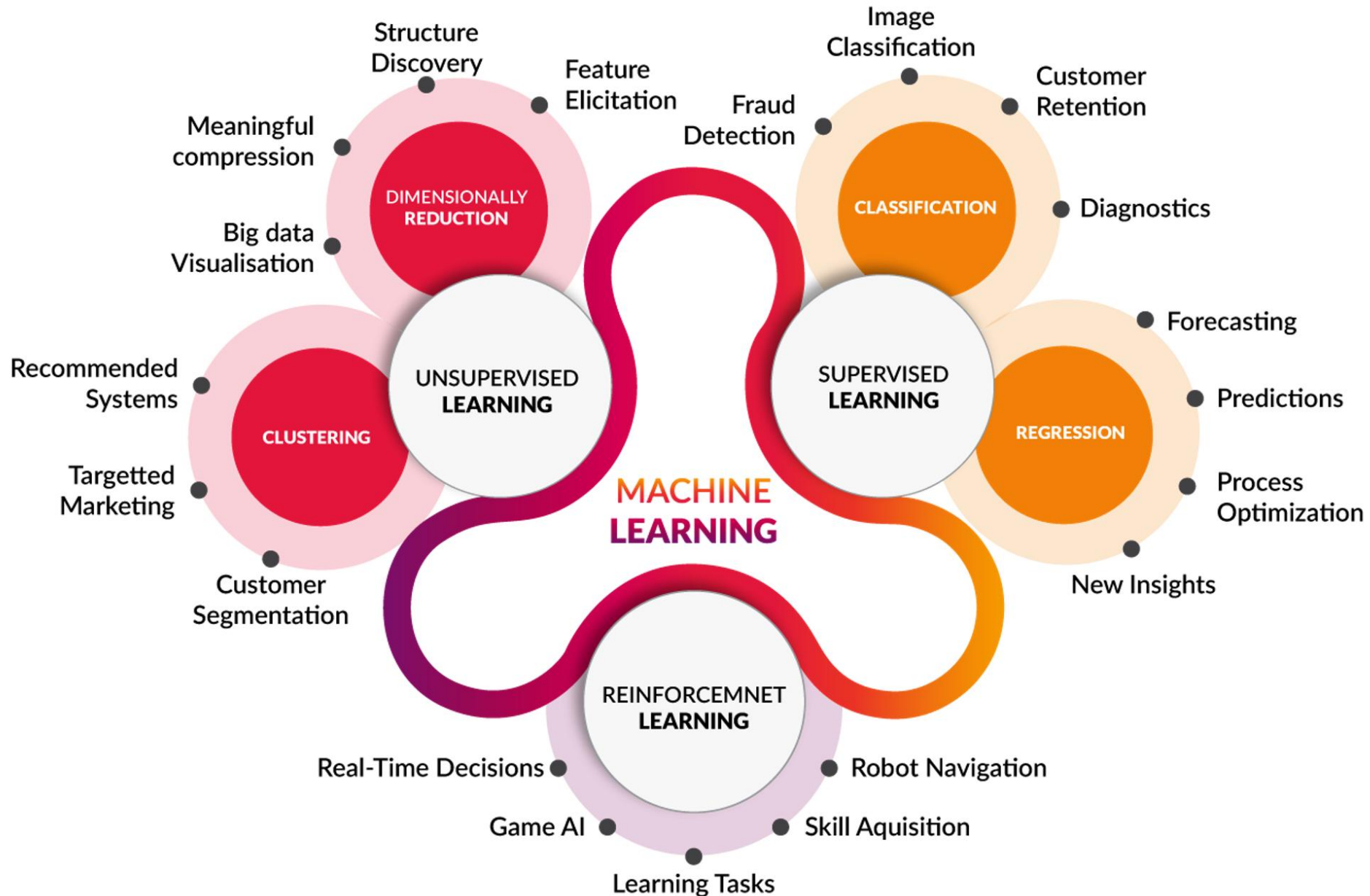


A common framework?

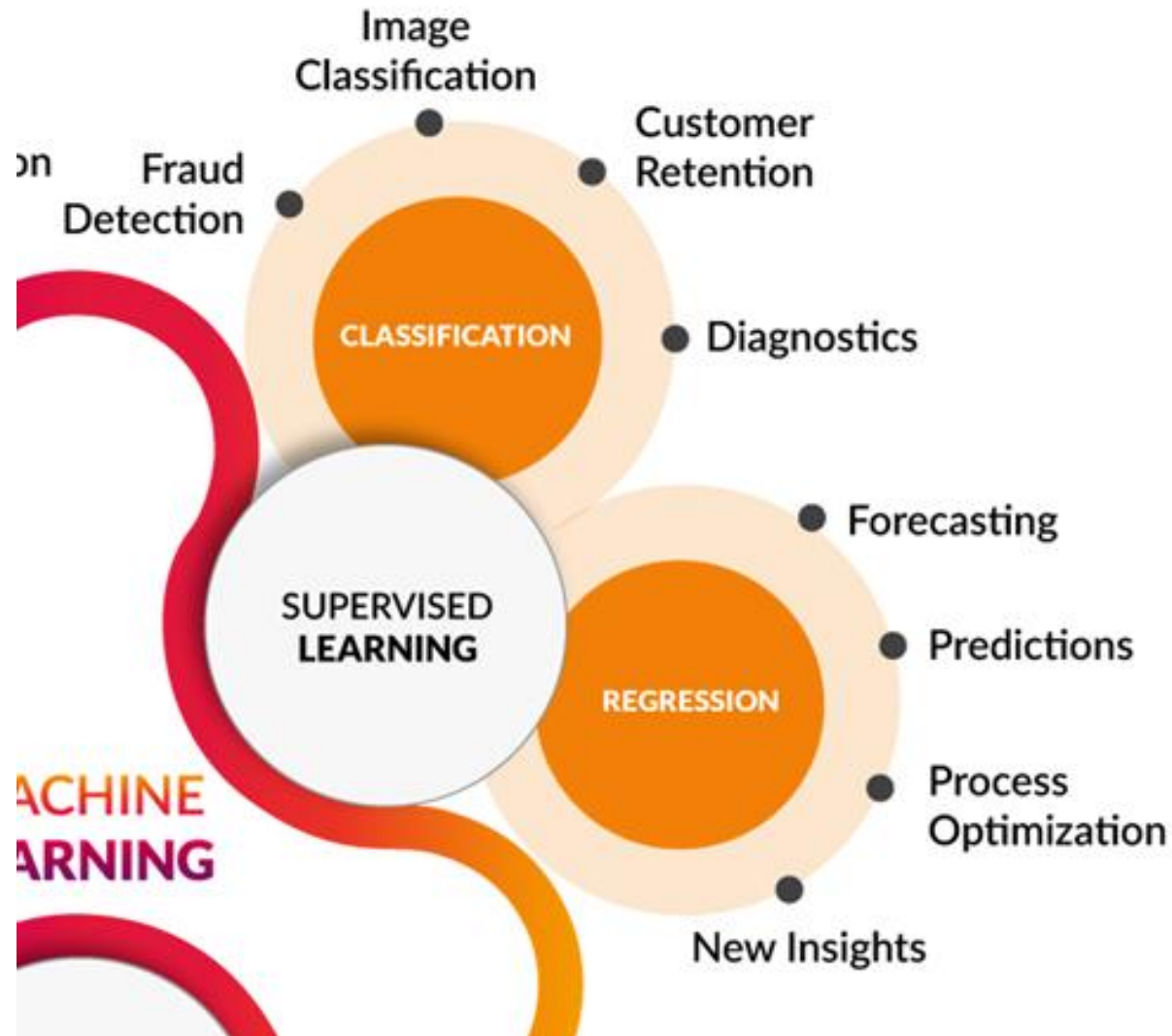
- So many topics, applications, fields...
- Is there some clever way to **group** and to **cluster** each application of the AI?
- Yes!
 - Let's study a general taxonomy for AI



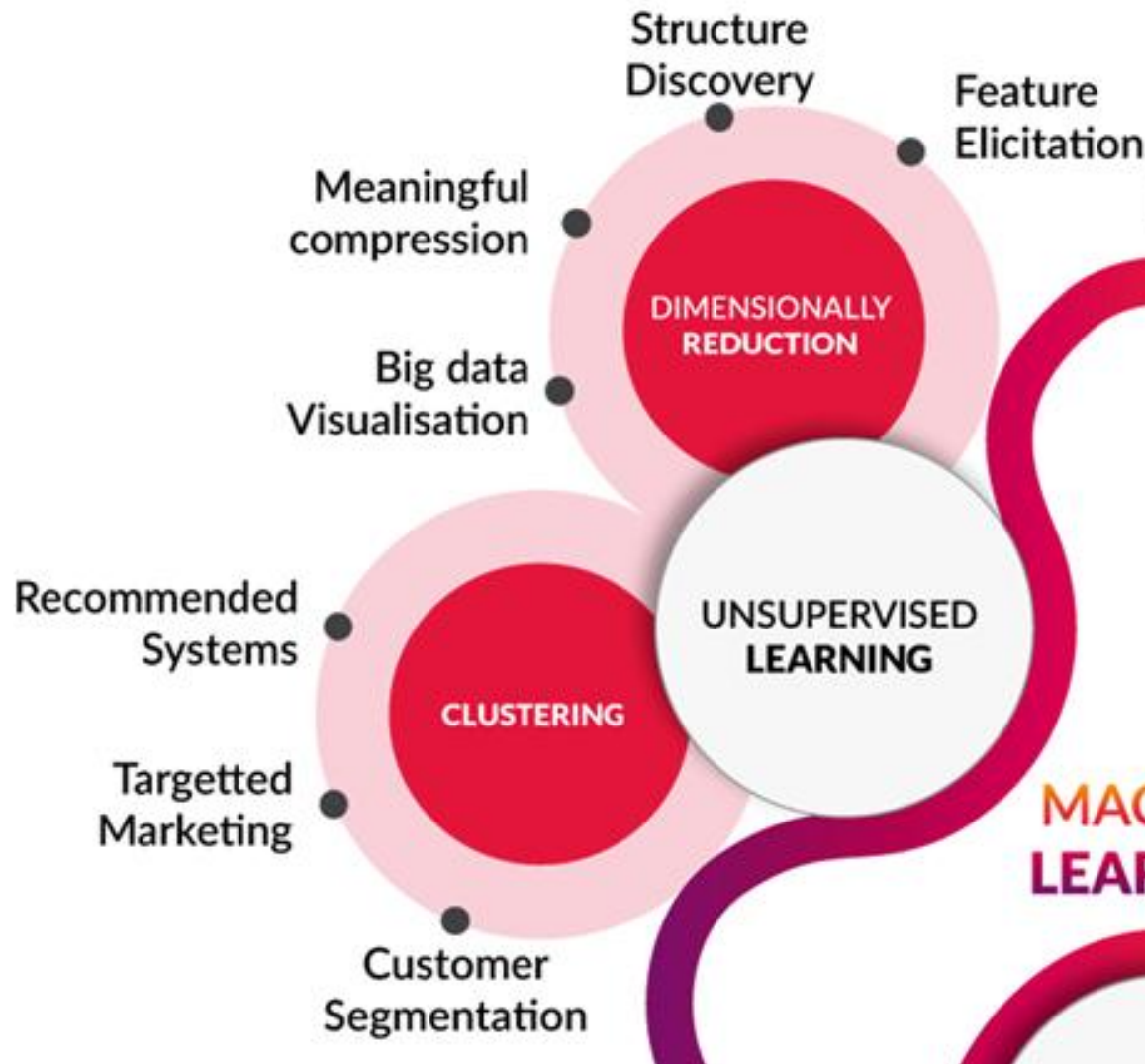
ML taxonomy



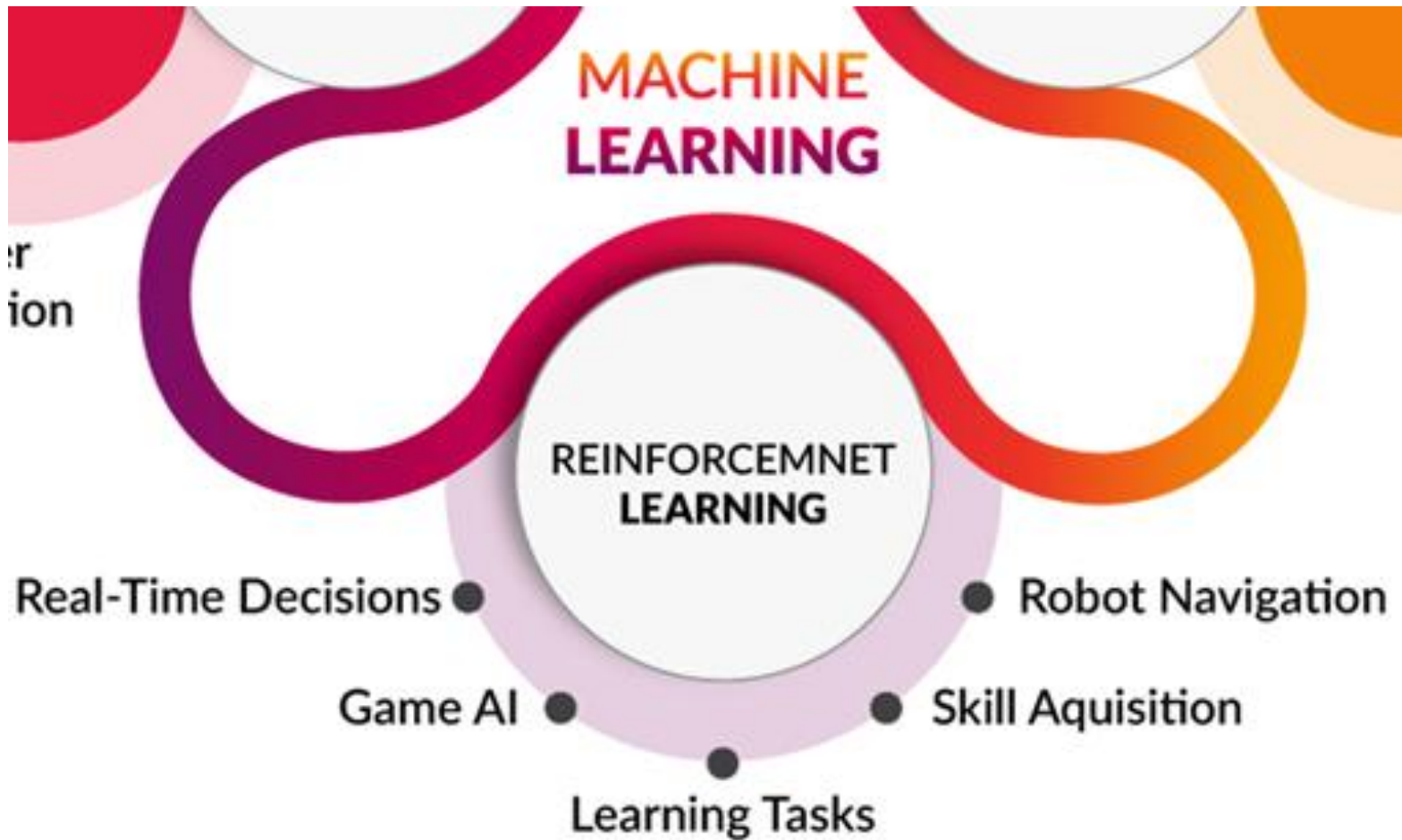
ML taxonomy



ML taxonomy

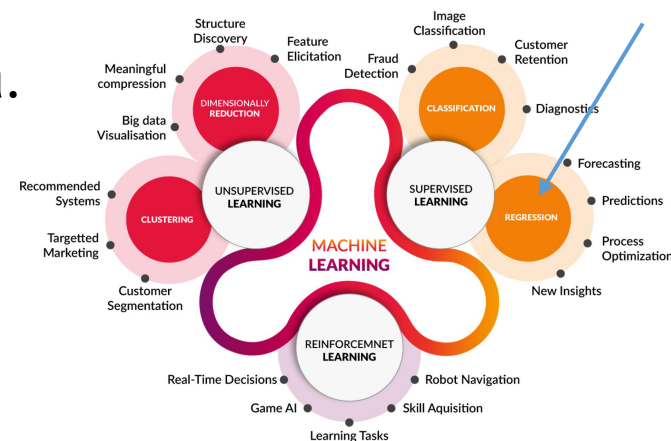
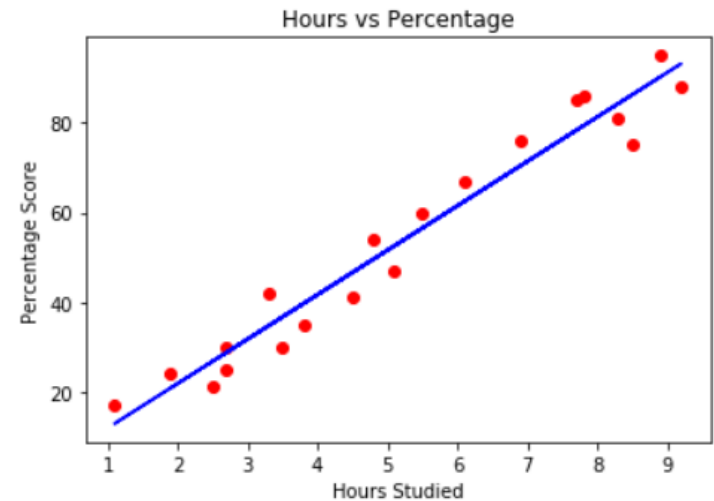


ML taxonomy



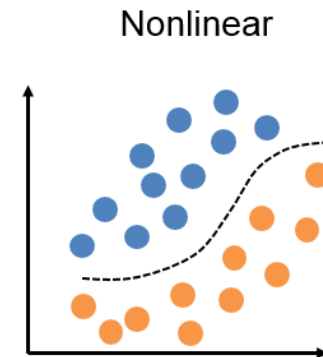
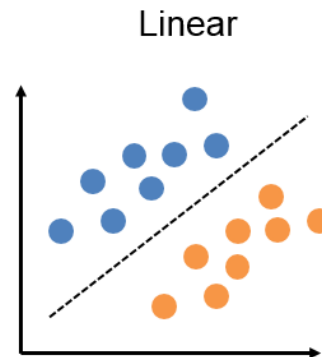
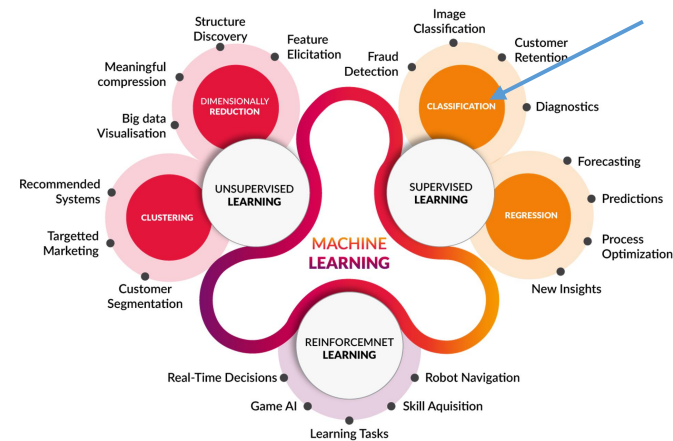
Regression

- It predicts **continuous valued output**.
- Compute the probabilistic relationship between *variables* for the purposes of *forecasting* or *prediction*.
- The Regression analysis is the statistical model which is used to predict the numeric data instead of labels.
 - That regression is the problem of predicting a continuous quantity output for an example
- It can also identify trends based on the available data or historic data.
- Example: predicting a person's income from their age, education, etc...

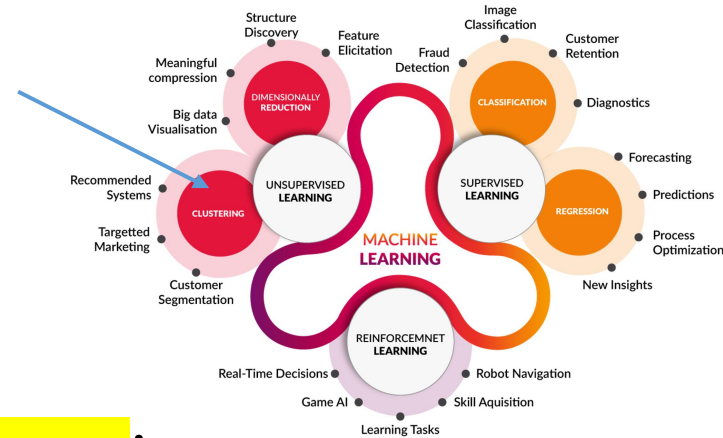


Classification

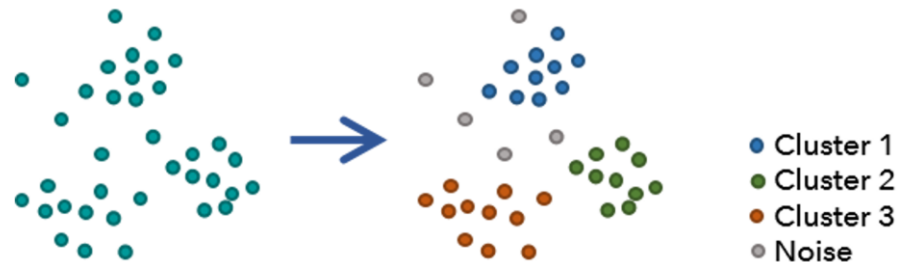
- It predicts **discrete number** of values processing the inputs
- In classification the data is categorized under different labels according to some parameters and then the labels are predicted for the data
- Classifying emails as either spam or not spam is example of classification problem
- Ex.: 2D input class.



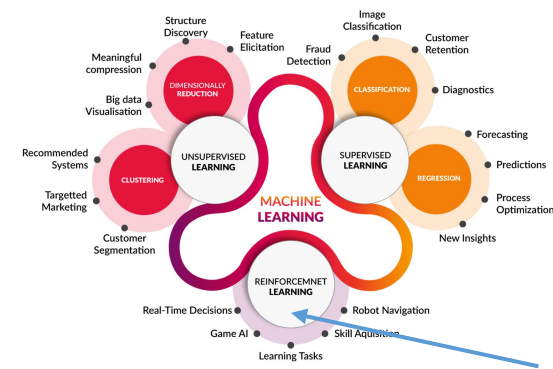
Clustering



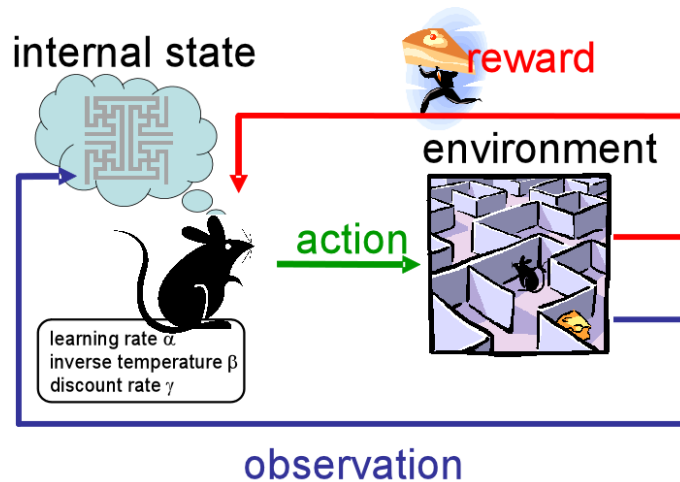
- the task of partitioning the dataset into groups, called clusters of similar items
- The goal is to split up the data in such a way that points within single cluster are very similar and points in different clusters are different
- It determines grouping among unlabeled data
- Ex.: 2D points



Reinforcement Learning



- Reinforcement learning models use opposite dynamics such as rewards and punishment to “reinforce” different types of knowledge.
- This type of learning technique is becoming really popular in modern AI solutions.



For example, the ability to develop behaviors/patterns such as:

- Random walk
- Go as straight as possible
- Spiral/whorl

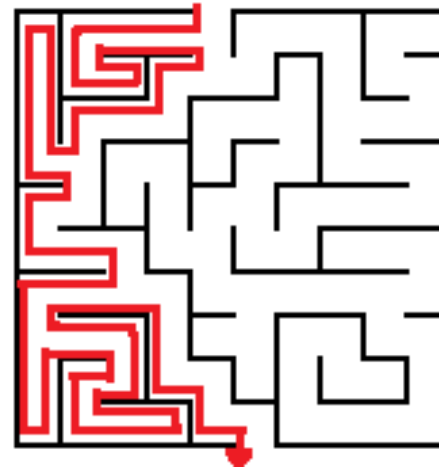
Reinforcement Learning.

- Different policies



Policy: follow rows in the room

Reward: with the same battery clear a larger surface.



Policy: Keep the hand on the wall to exit the maze.

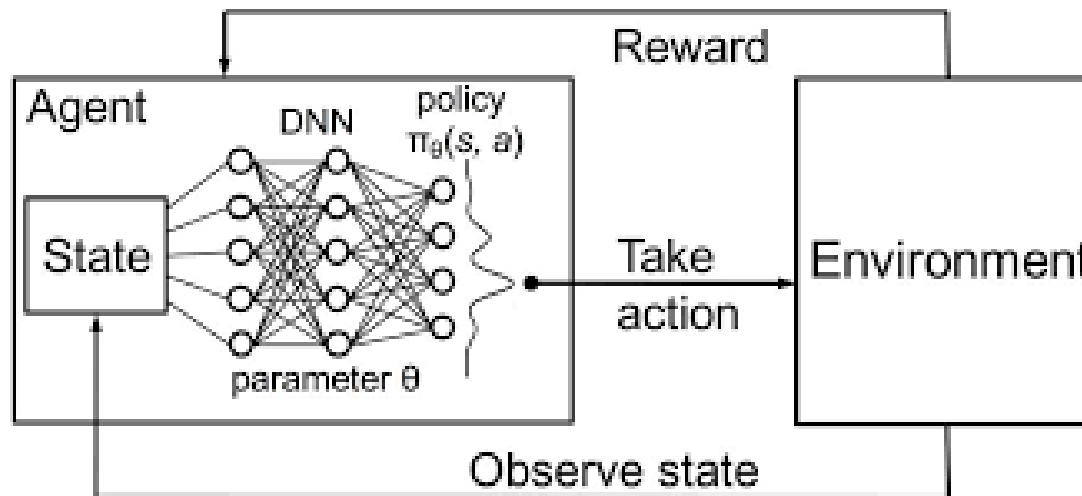
(Better than make random turns...)

Reward: exit the maze, with less steps



Reinforcement Learning..

- Goal: get as much reward as possible!

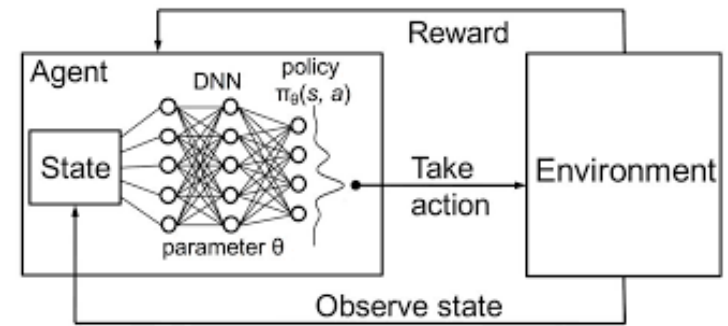


- The neural network output is a probability distribution over possible actions.
- Example: If the agent can take three actions A_1, A_2, A_3 , the policy might return:

$$\pi_{\theta}(s, A_1) = 0.2, \quad \pi_{\theta}(s, A_2) = 0.5, \quad \pi_{\theta}(s, A_3) = 0.3$$

- The agent selects an action by sampling from this distribution.

Reinforcement Learning...



Agent: The entity making decisions to maximize a reward.

State (s): The representation of the current condition of the environment, which the agent observes.

Deep Neural Network (DNN): The agent uses a neural network to approximate a policy (π) that maps states to actions.

Policy ($\pi_{\theta}(s, a)$): A function that dictates the probability of choosing action a given state s , where θ are the network parameters.

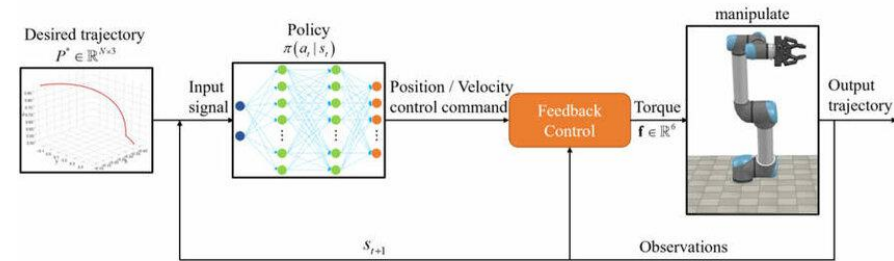
Action: The decision or movement the agent takes based on the policy.

Environment: The external system where the agent operates and takes actions.

Reward: The feedback signal given to the agent based on its action, guiding learning.

Observation (Observe State): The agent receives the new state from the environment after executing an action.

Reinforcement Learning: the robot arm



Problem Setup

- We have a **robotic arm** with multiple joints.
- The goal is to **grasp an object** placed at a random position on a table.
- The arm is controlled by **torques applied to its joints**.
- The agent (robot) **receives rewards** based on how well it approaches and grasps the object.

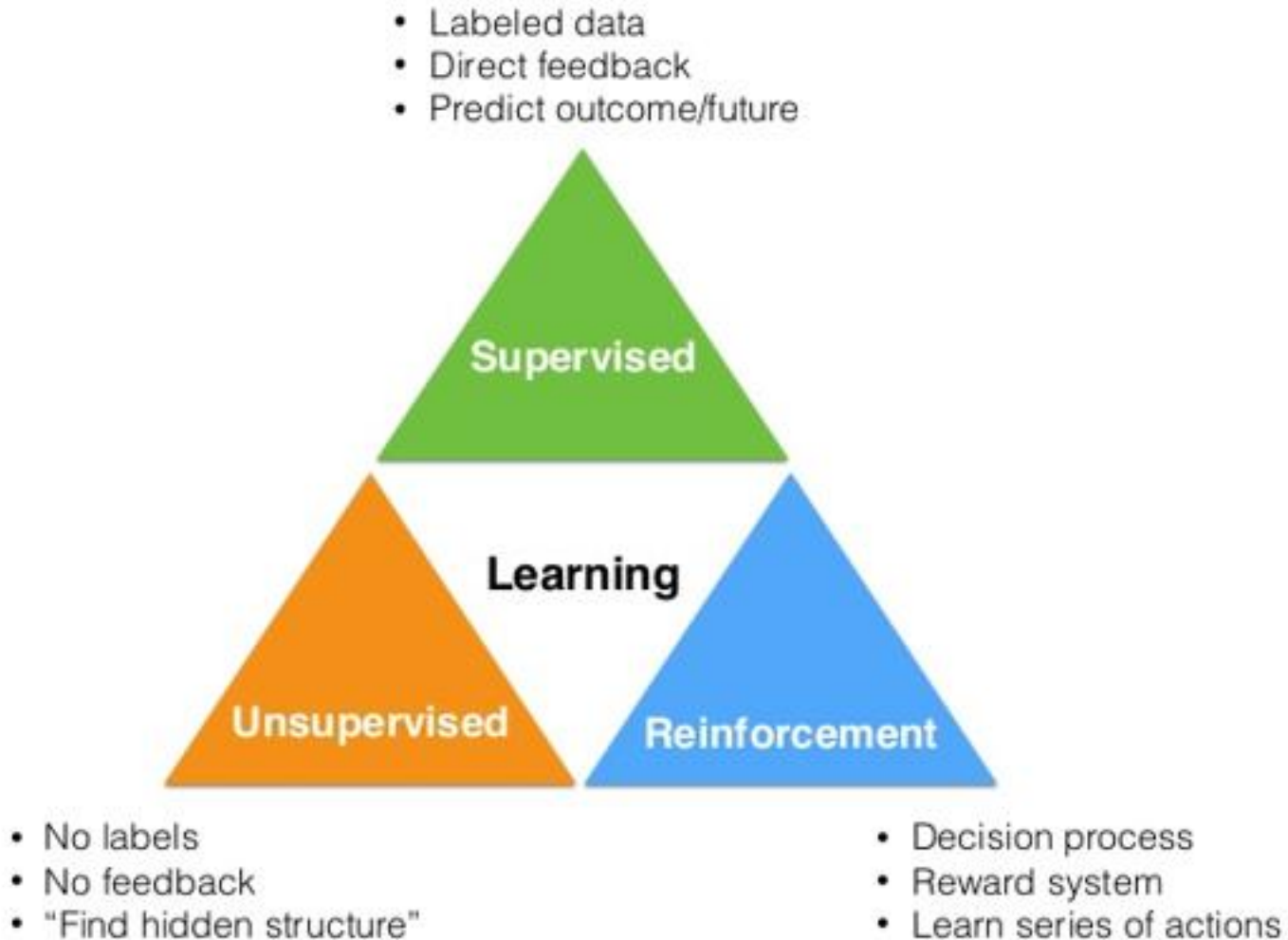
Reinforcement Learning Components

1. **State (s):** The state represents the **current position and orientation** of the robot's end-effector (the gripper) and the object's position (e.g., joint angles and velocities, 3D position of the object, distance between the gripper and the object)
2. **Action (a):** The action space consists of the **torques applied to each joint**.
3. **Policy ($\pi(\theta(s,a))$):**
 1. If using a **stochastic policy**, the output is a probability distribution over different torque values.
 2. If using a **deterministic policy**, the output is the exact torque to apply.
4. **Reward Function (R):** The reward function guides the learning process. It may include:
Positive rewards: +10 if the robot successfully grasps the object., +1 if the end-effector moves closer to the object.
Negative rewards: -1 for excessive torque usage (to encourage efficiency), -5 if the robot drops the object.
5. **Environment:** A physics simulator like **PyBullet, MuJoCo, or Gazebo**

Types of Learning VS labels

- **Supervised** (inductive)
 - Training data includes desired outputs (labels)
- **Unsupervised**
 - Training data does not include desired outputs
- **Semi-supervised**
 - Training data includes a few desired outputs
- **Reinforcement**
 - Rewards from sequence of actions

Types of Learning: another point of view

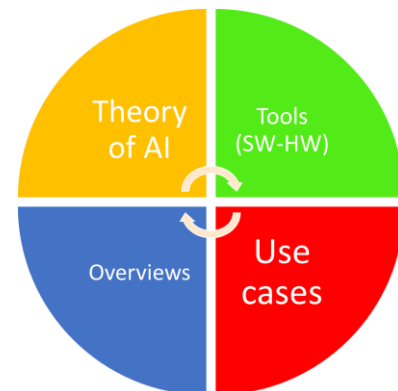




THEORY

Deductive and Inductive learning

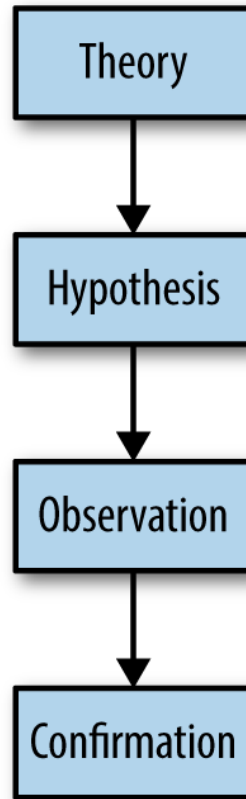
Learning from data or applying rules?



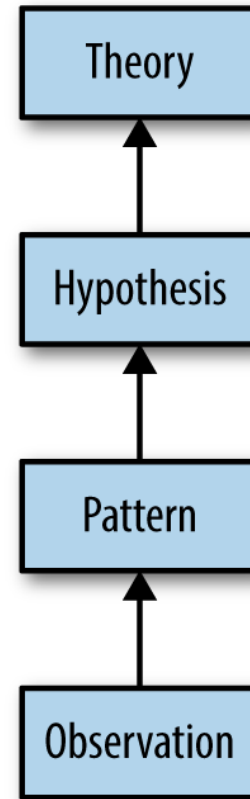
Deductive VS Inductive

Example:
If-then
rules,
no data

Deductive Reasoning



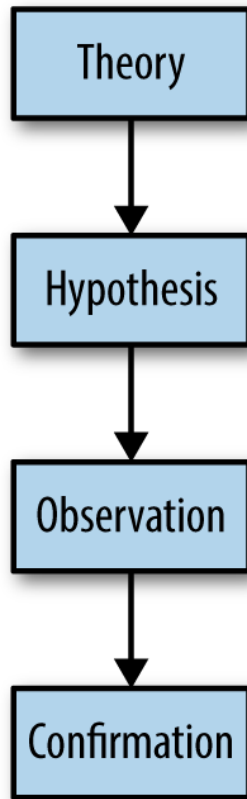
Inductive Reasoning



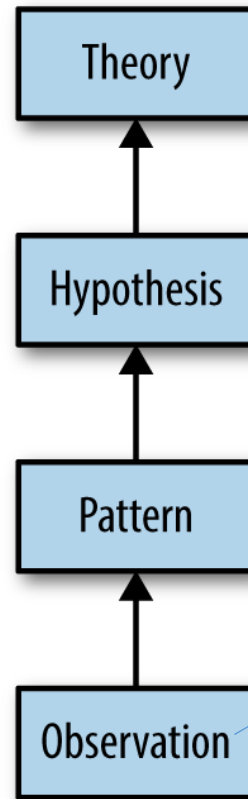
Deductive VS Inductive

Deductive Reasoning

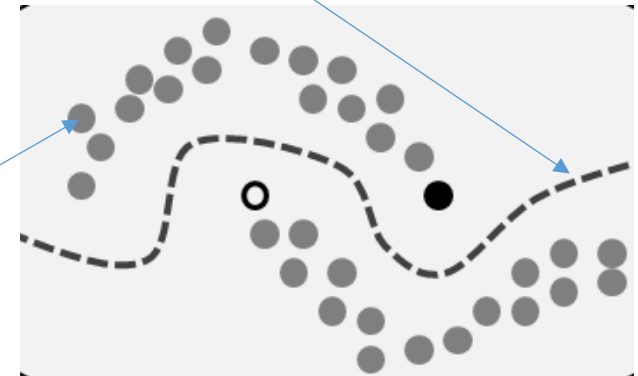
If-then
rules,
no data



Inductive Reasoning



No previous knowledge,
just data



Deductive learning

"domain expert passing down rules"



Let's use a mnemonic...
your grandmother is giving you the
rules to choose a good fiancé

(if you agree...) you are inheriting
knowledge from her previous
experience and to apply them
to choose the new partner

Then... You can also adapt the rules
to modern time.

Deductive learning is a type of AI technique starting with a set of rules
and process new decisions on new data

Then... inferring new rules that are more efficient in the context to be applied

Inductive learning

"Learner without a prior"



James Dean, Rebel Without a Cause (1955)

Let's use a mnemonic...
you don't want rules and
you want gather your
knowledge by experience
(only data! No rules!)

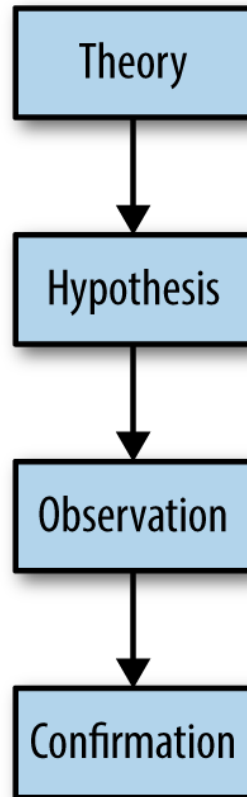
Last 10 times I entered Milano's highway
@17:30 I got stuck in a jam!

→ NEW GENERAL RULE: Avoid departure
in heavy traffic time

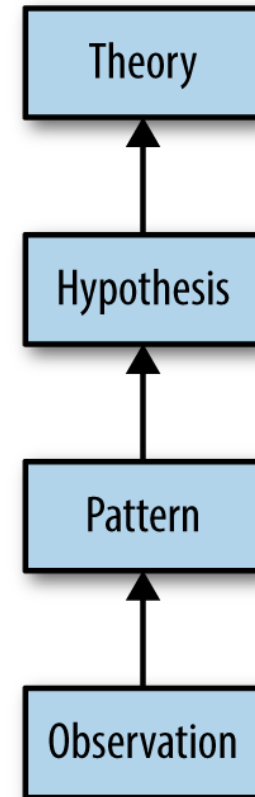
Inductive learning is based on inferring a general rule from datasets of input-output pairs

Deductive VS Inductive

Deductive Reasoning



Inductive Reasoning



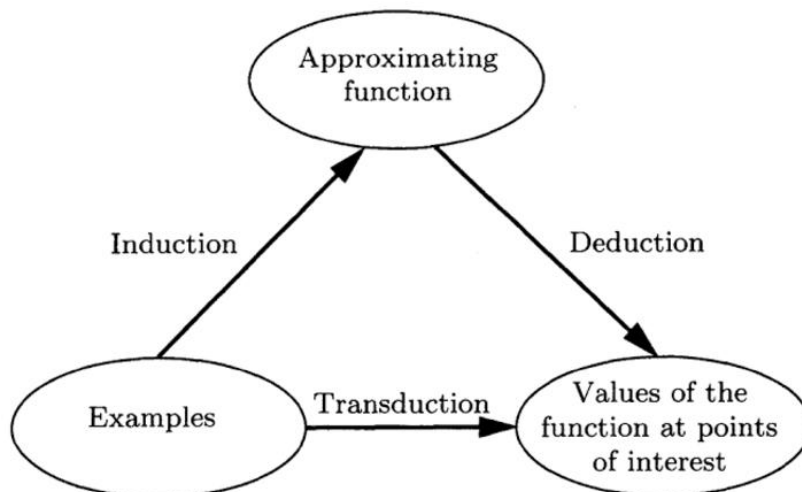
No previous knowledge,
just data

**THE MAJORITY
OF AI METHODS**



And Transduction...

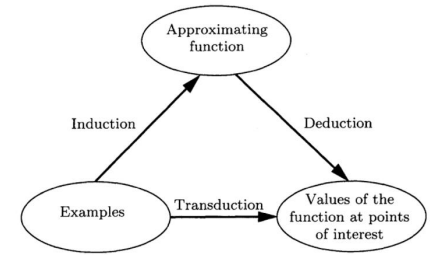
- **Induction**, deriving the function from the given **data**
- **Deduction**, deriving the values of the given **function** for points of interest
- **Transduction**, deriving the values of the unknown function for **points** of interest from the given data



The Nature of Statistical Learning Theory

Many natural language processing (NLP) tasks can be viewed as transduction problems since the model converts one string into another.

Transduction



- Moving from the particular (datapoints) to the particular (datapoints)
- Examples
 - **k-Nearest Neighbors:** an algorithm that does not model the training data, but instead uses it directly each time a prediction is required
 - **Linguistics:** a set of rules for transforming examples of one language into another
 - **Sequence Prediction:** a model that outputs one time step for each input time step provided (e.g., Recurrent Neural Networks)

Inductive Learning

- Given examples of a function $(X, F(X))$

- $X=0,2 \rightarrow Y=32;$

- $X=0,5 \rightarrow Y=52;$

-

- (1-Dimensional case)



$\rightarrow Y= 1;$



$\rightarrow Y= 1;$



$\rightarrow Y= 2;$

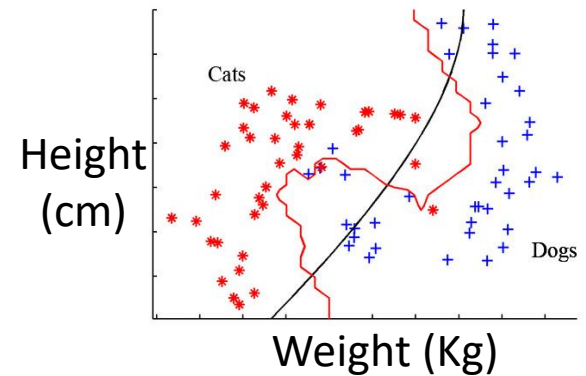
(N-Dimensional case)

Inductive Learning (2)

- Create a prediction function $F(X)$ for new examples X

- Discrete $F(X)$:

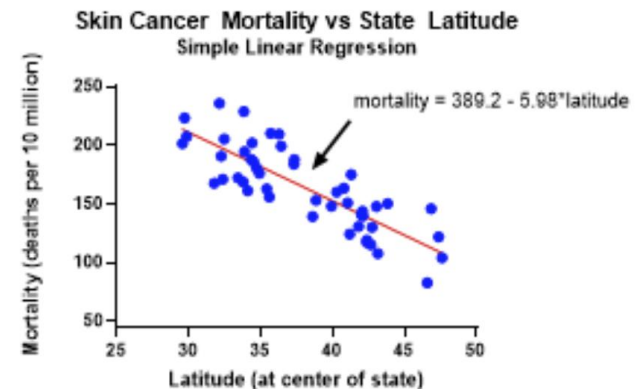
- **Classification** $\text{out}=F(X)$ is an *integer*



- Continuous $F(X)$:

- **Regression**

- Example:
 $F(X) = \text{Probability}(X)$
Probability estimation



Inductive Learning:

Advantages

- Alleviate knowledge acquisition Bottleneck (no rules, no equations, just data)
 - Does not require knowledge engineers
 - Scalable in constructing knowledge base
- Adaptive
 - Adaptive to the changing conditions
 - Easy in migrating to new domains

Inductive Learning: Generalization

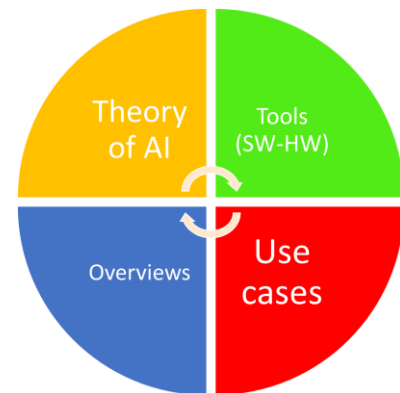
- The obtained (trained) model can generalize the capability to solve a problem in the specific application to **more general cases**
- IF (there are a lot of IFs)
 - the dataset is good and complete,
 - the learning method and the model are enough powerful
 - etc..



THEORY

Evaluation and Assessment of an AI System

Is my model ok for my application?



Evaluation and Assessment of an AI System

- This a very complex and articulate field of research
- The following aspects will be detailed in the course
- **Objective:** Define the purpose of evaluation
 - Ensure the AI system meets performance expectations
 - Guarantee safety and reliability
 - Identify areas for improvement

Evaluation and Assessment of an AI System

Metrics: Quantitative and Qualitative Measures

- Accuracy, Precision, Recall, F1 Score
- Area Under the Curve (AUC), Receiver Operating Characteristic (ROC)
- Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE)
- User satisfaction and usability
 - Ergonomics,
 - In the future even the user experience will be evaluated!
Is my AI model empathic? Assertive? Nice? Verbose?.....
- Ethical considerations and fairness
 - Created with biased data?

Evaluation and Assessment of an AI System

Validation Methods: Techniques to Test AI Models

- Training/Validation/Test Split
- Cross-Validation
- Bootstrap
- In the future, new possible test like User-experience based test? Psychological-like test?

Evaluation and Assessment of an AI System (cont.)

- **Benchmarking:** Comparison with Baselines
 - Comparison with previous models or human performance
 - Industry standards and benchmarks
 - Public leaderboards
- **Real-world Performance:** Beyond the Numbers
 - User studies and field tests
 - Robustness and generalizability assessments
 - Stress testing

Evaluation and Assessment of an AI System (cont.)

- **Continuous Monitoring:** Ensuring Long-Term Reliability
 - Data drift detection
 - Model retraining strategies
 - Performance degradation alerts
- **Ethics and Compliance:** Meeting Legal and Societal Standards
 - Bias and fairness analysis
 - Privacy and security evaluation
 - Regulatory compliance checks (AI Act 2024)

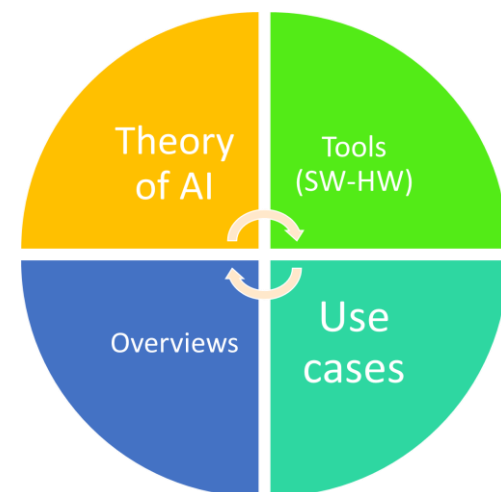




Overview

Intelligent sensors,
and environmental control

Virtual sensors



Intelligent sensors

- Heterogeneous multi-sensor systems.
- Sensor data analysis.
- Diagnosis.
- Fault tolerance.
- Self-calibration.
- Adaptivity.
- Management.

Example

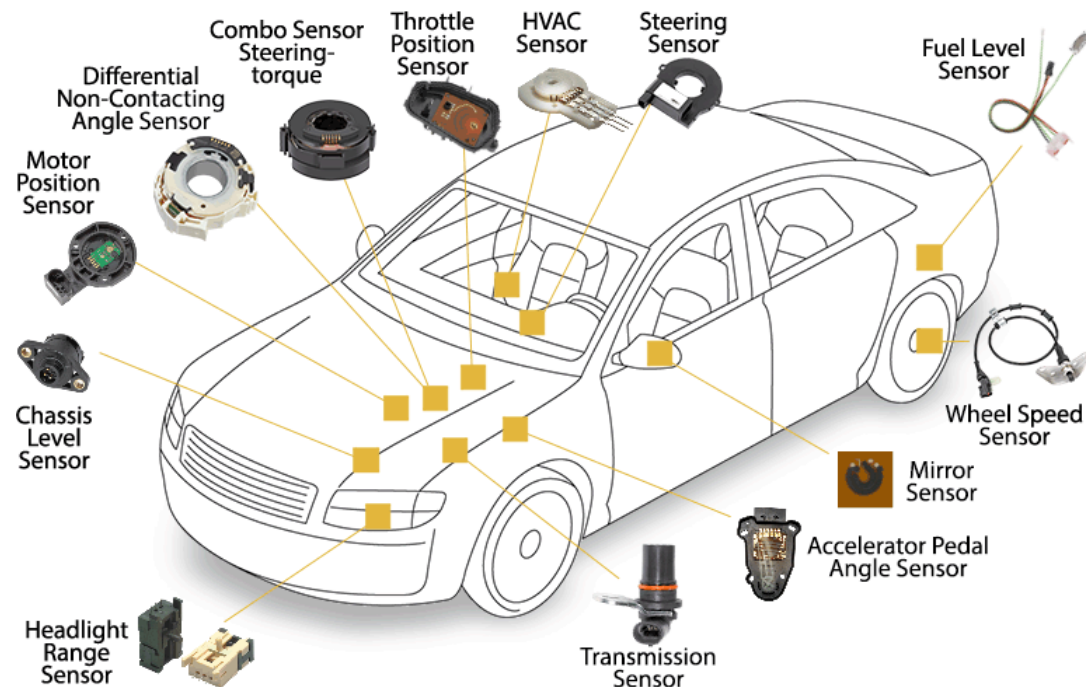
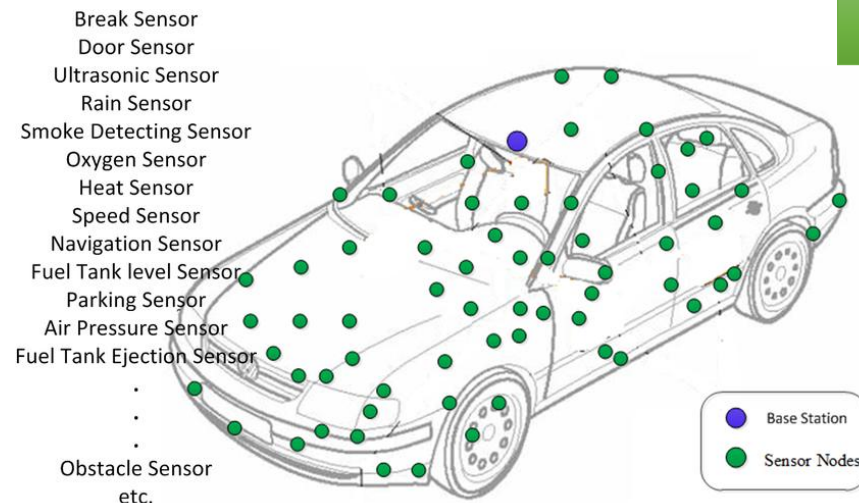


Nest Labs: co-founded Tony Fadell and Matt Rogers in 2010 (former Apple engineers); the flagship product was the Nest Learning Thermostat (from 2011).

Google acquired Nest Labs for **3.2 billion US\$** in January 2014 (with 280 employees).

Sensor networks

- Structure
- Functions
- Adaptivity
- Management
- Distributed data analysis
- Fault tolerance
- Diagnosis



Instrumentation and Measurements



- Acquisition and processing of sensor measurement in advanced adaptive infrastructures
- Multi-sensorial data fusion
- AI-based **Virtual Sensors**
 - AI provides feasible and economical alternatives to costly or impractical physical measurement instrument.
A virtual sensing system uses information available from other measurements and process parameters to calculate an estimate of the quantity of interest

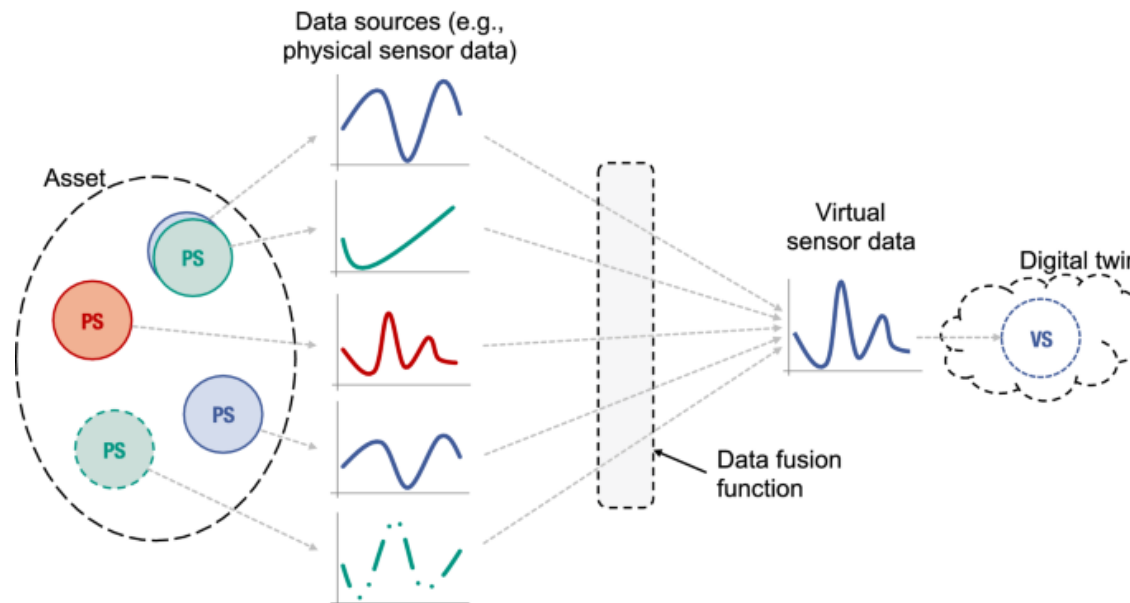
Virtual Sensors

What is a Virtual Sensor?

- A software-based system that estimates a physical quantity using data from multiple real sensors and computational models.
- Replaces or complements physical sensors when direct measurement is expensive, impractical, or impossible.

AI-Based Virtual Sensor Workflow

1. Data Collection and harmonization
2. Feature Extraction
3. Modeling & Estimation
4. Validation & Refinement



Virtual Sensors



Example: AI-powered virtual sensor that estimates the **SOC** (State Of Charge) using inputs such as **voltage, current, and temperature**

Key Advantages

- Cost reduction (fewer physical sensors required)
- Increased reliability (redundancy and error correction)
- Enhanced data fusion (integration of multiple sources)
- Improved predictive capabilities (ML-based estimations)

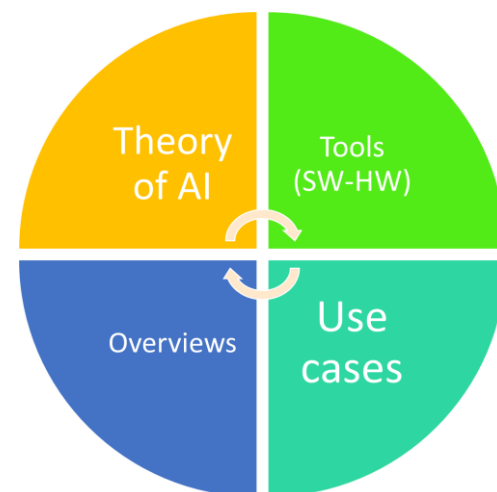
Example Applications

- Industrial process monitoring (estimating gas flow, temperature)
- Automotive industry (virtual speed and tire pressure monitoring)
- Environmental monitoring (air quality, water contamination detection)
- Predictive maintenance in Industry 4.0



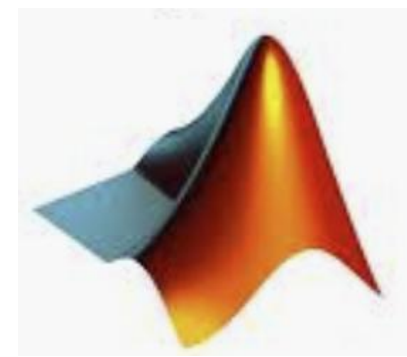
Toolboxes

Toolboxes to **prepare**
for the next lessons



TOOLS

- Few notes just to **PREPARE** the toolboxes for the next lesson.
- We will review in detail the tools in the next lessons
- Main tools used in the course
 - **Matlab** (many many many tools and functions)
 - **Colab** (Online servers, TPUs, ...)
 - **Keras** (The Python Deep Learning library)
- They offer the best tradeoff between *usability* and *complexity*, and a good learning curve.
- Supporting Keras and many NN pretrained models.

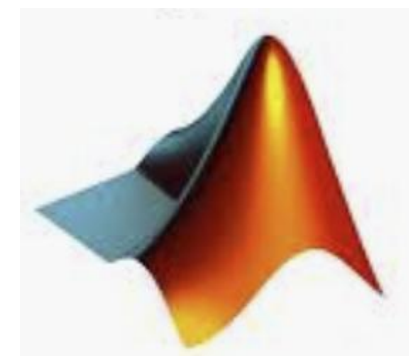


TOOL#1: Matlab

- Matlab is the first tool in our course
 - Not only machine learning but much much more.
 - Thousands of well-documented and powerful functions
- Read the instruction and download your free license of Matlab as UNIMI student
- Download the **last** version! The toolboxes are improving fast!
- https://work.unimi.it/servizi/servizi_tec/79539.htm
(or just Google>> UNIMI MATLAB)
- See instruction to install MATLAB on an **offline machine**.

TOOL#1:

Matlab Toolboxes



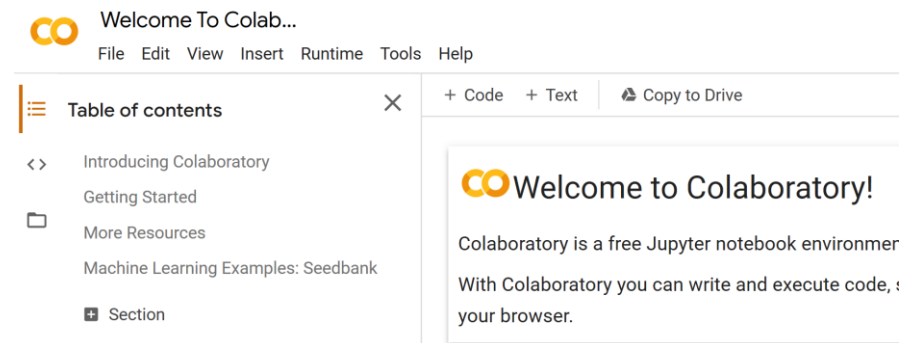
- Install all toolboxes, but if you are short of space on your PC HD / reduced bandwidth, install just the **main toolboxes** needed for the course
 - Deep Learning Toolbox
 - Fuzzy Logic Toolbox
 - Statistics and Machine Learning Toolbox
 - Image Processing Toolbox
 - Signal Processing Toolbox
 - Computer Vision Toolbox
 - Choose some toolbox to be used in the future for your projects, it's free!....

<https://it.mathworks.com/products/deep-learning.html>

TOOL#2:

Google Colab

- COLAB=Colaboratory is a free **Jupyter notebook** environment that requires no setup and runs entirely in the cloud.
- You can
 - write and execute code,
 - save and share your analyses,
 - and access powerful computing resources,
 - all for free from your browser.



TOOL#2:

Google Colab



Google

Sign in

Use your Google Account

Email or phone

- COLAB
 - It is very very useful!
 - It helps us to optimize the time in the course, reaching our goals without wasting time in technicalities
 - But since it requires a google account to save data and code it is **not** strictly compulsory for the course
- You have other options (less easy and ready-to-use)
 - use Matlab instead for all
 - download other local tools (you probably need GPUs) like Jupyter notebook, Ananconda, python, etc.
 - you can create/ use a specific Google Account without any personal information stored in it
 - **Do not ask for assistance...**

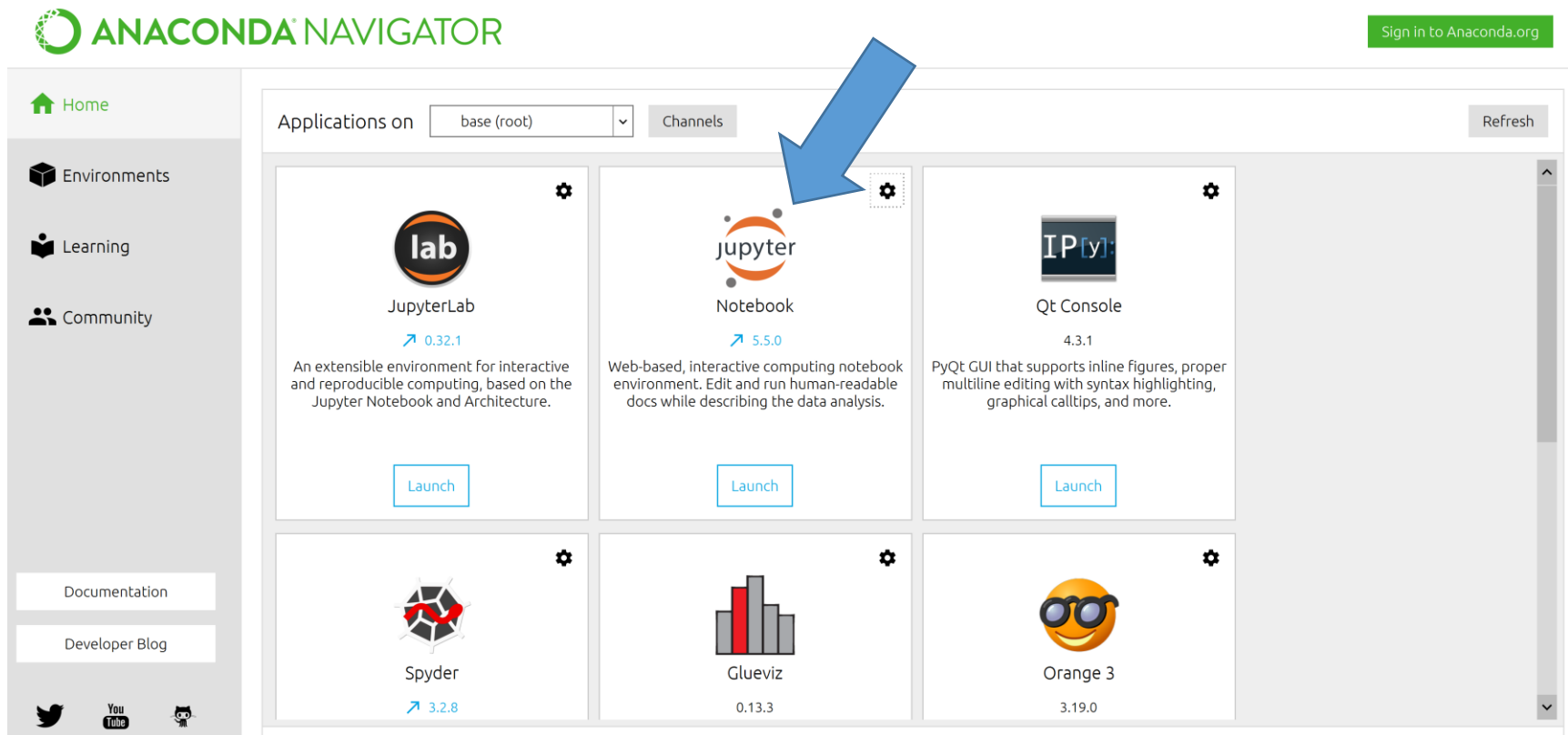
TOOL#3



- A powerful tool for **scientific computing and data analysis** that simplifies package management and deployment, making it a popular choice among data scientists, researchers, and developers
- Distribution of the **Python and R**.
- **Many pre-installed libraries** and tools that are commonly used in data science, machine learning, and other scientific applications
- Very good **package management system** (ease install, update, and manage different versions of packages and dependencies)

Anaconda Navigator

Desktop application lets you easily manage integrated applications, packages, and environments without using the command line.



Jupyter Notebook



localhost:8888/notebooks/Untitled.ipynb

jupyter Untitled Last Checkpoint: alcuni secondi fa (autosaved)

File Edit View Insert Cell Kernel Widgets Help

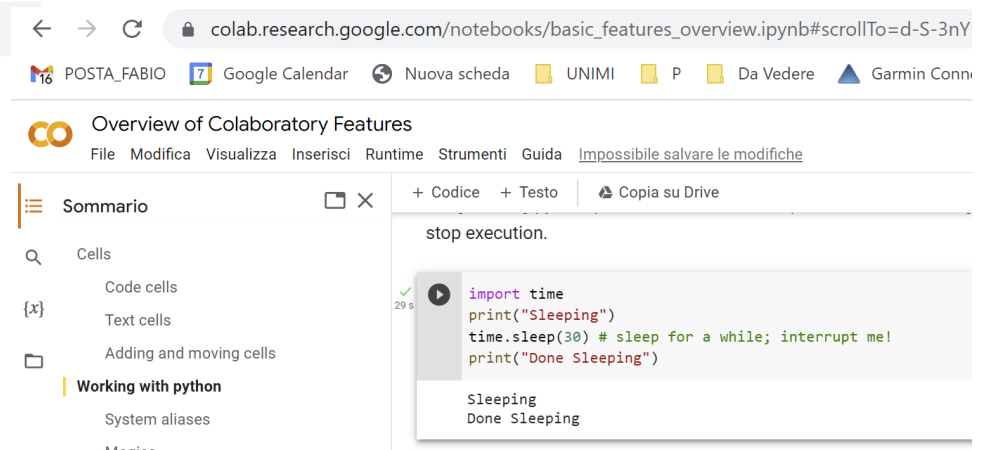
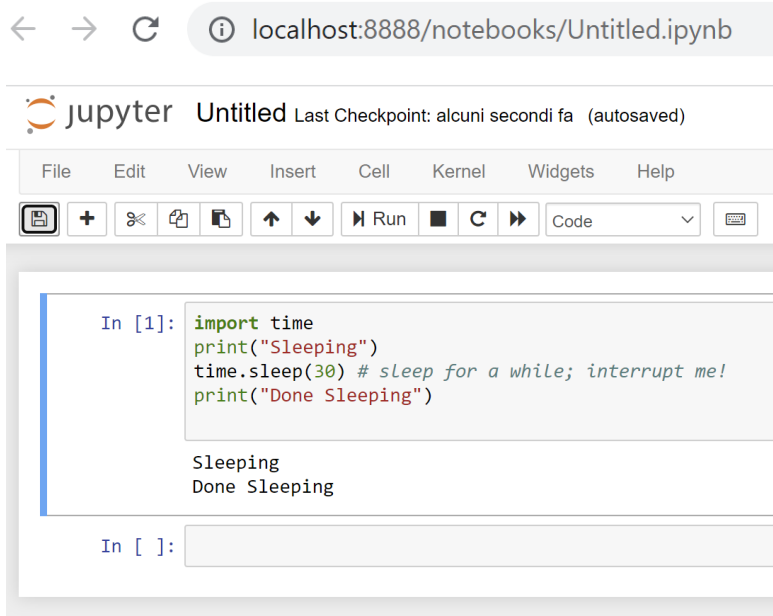


```
In [1]: import time
        print("Sleeping")
        time.sleep(30) # sleep for a while; interrupt me!
        print("Done Sleeping")
```

Sleeping
Done Sleeping

In []:

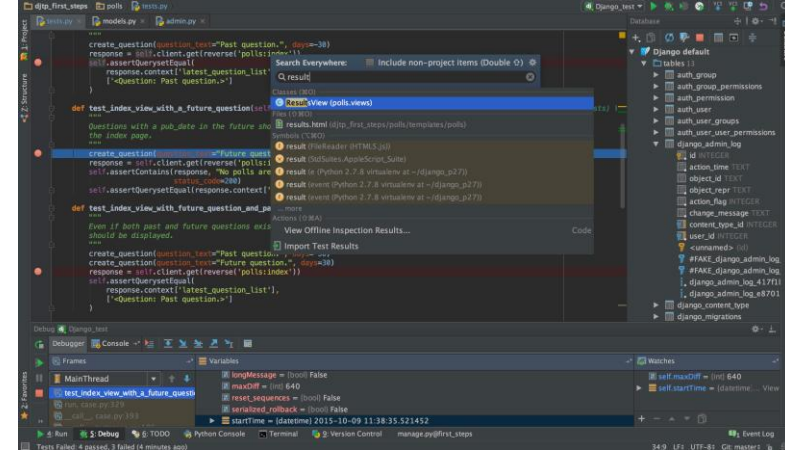
Jupyter Notebook and Colab



Google Colab is ready and more hardware-independent (e.g., test your code on TPUs etc..) with some limitations which are not so relevant in this course

Or use an IDE

(Integrated Development Environment)



PyCharm

The Python IDE
for Professional Developers

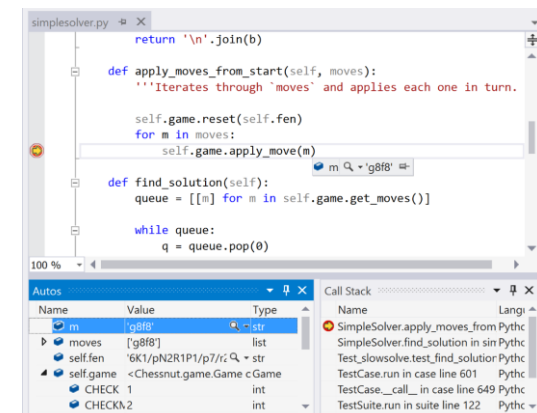
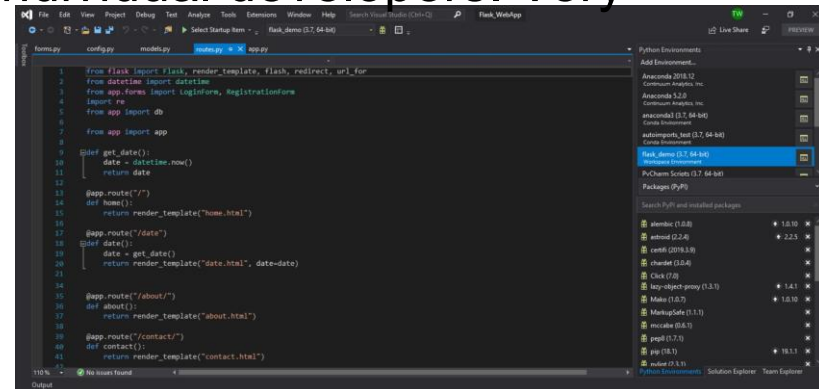
JET
BRAINS

IDE: MS Visual Studio (Community)

There is a free version of Visual Studio available for students: **Visual Studio Community**. This is a fully-featured, extensible IDE (Integrated Development Environment) provided by Microsoft at no cost for students, open-source contributors, and individual developers. Very good debugging tools.

Tools which ideal to develop ISs in VS:

- GitHub Copilot,
- ML.NET Model Builder
- IntelliCode,
- Azure Machine Learning Extension,
- TensorFlow.NET Tools, Bito (An AI coding assistant alternative to Copilot),
- Python Tools for Visual Studio (PTVS) (An open-source extension that turns Visual Studio into a Python IDE, with NumPy, Pandas, and scikit-learn)

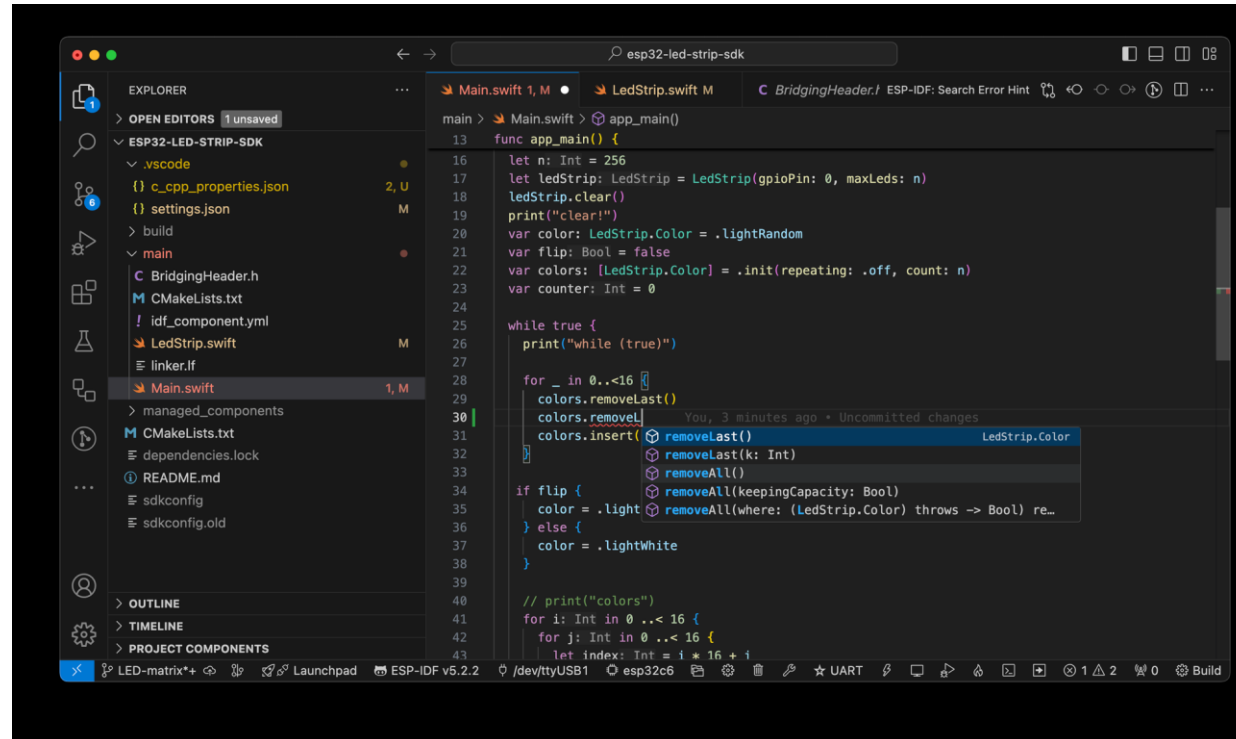


Visual Studio Code

Visual Studio Code is a free, lightweight and flexible IDE with strong support for Python, extensions, and AI development workflows.

Github,
Extensions,
Coding agents,
Copilot, etc.

All in one click!



A final comparison

(not in the exam...)

Tool	Ease of Use	Simplicity of Installation	Extensibility
MATLAB (Free for Students)	High: Intuitive GUI, built-in toolboxes (e.g., Deep Learning Toolbox), and extensive documentation make it beginner-friendly for	Moderate: Requires institutional license setup (assumed free via your school), but installation can be complex for novices due to size (~3-5 GB) and configuration.	Moderate: Rich ecosystem of toolboxes (e.g., Statistics, Neural Networks), but limited to MATLAB's proprietary environment; less flexible for non-MATLAB libraries.
Google Colab	Very High: Cloud-based, no setup needed, pre-installed AI libraries (e.g., TensorFlow, PyTorch), and GPU/TPU access simplify AI prototyping.	Very High: No installation required—just a browser and Google account. Ideal for instant use.	Moderate: Extensible via pip/conda installs within notebooks, integrates with Google Drive, but limited to Python/R and cloud constraints (e.g., session timeouts).
Anaconda with Jupyter Notebook	High: Jupyter's notebook interface is interactive and great for data exploration/AI experimentation; Anaconda simplifies package management.	Moderate: One installer (~500 MB) bundles Python, Jupyter, and AI libraries (e.g., NumPy, Pandas), but requires initial setup and updates.	High: Highly extensible with conda/pip for thousands of packages, supports multiple languages (Python, R, Julia), and integrates with extensions (e.g., JupyterLab).
PyCharm	Moderate: Feature-rich IDE with code completion and debugging, but steeper learning curve for beginners compared	Moderate: Straightforward download (~600 MB), but requires configuring interpreters (e.g., Python, Anaconda) and plugins for AI tools.	Very High: Extensive plugin ecosystem (e.g., Git, Docker), supports AI frameworks (TensorFlow, PyTorch), and excels in large-scale project development.
Visual Studio Community	Moderate: Powerful IDE with good support for Python/C# AI tasks, but interface can overwhelm novices; Copilot integration simplifies coding.	Moderate: Large installer (~1-2 GB depending on workloads), easy for Windows users, but requires selecting AI-related components (e.g., Python development).	Very High: Supports multiple languages, integrates AI tools (e.g., ML.NET, TensorFlow.NET), and offers extensions (e.g., Azure ML, GitHub Copilot).

KERAS



Keras

- **The Python Deep Learning library**
- A high-level neural networks API, written in Python and capable of running on top of TensorFlow, CNTK, or Theano enabling fast experimentation.
- *From idea to result with the least possible delay*
- *No need to download, it's inside (Matlab, Colab)*

```
from keras.preprocessing import image
from keras.applications import resnet50

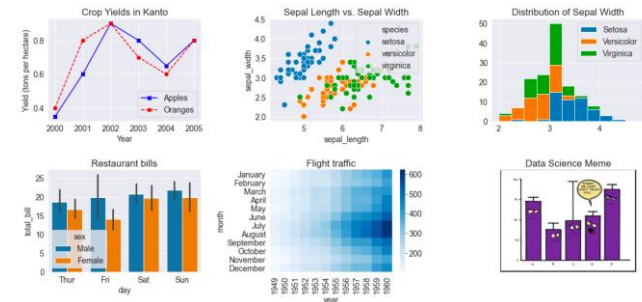
# Load Keras' ResNet50 model that was pre-trained
model = resnet50.ResNet50()

# Load the image file, resizing it to 224x224 pixels
img = image.load_img("bay.jpg", target_size=(224, 224))

# Convert the image to a numpy array
```

Class poll

- How many of you
 - Are able to code in Matlab or Python?
 - Already trained a neural network?
 - On his own data, not a standard dataset
 - Are using Colab?
- What library are you using for plots and graphical representation?

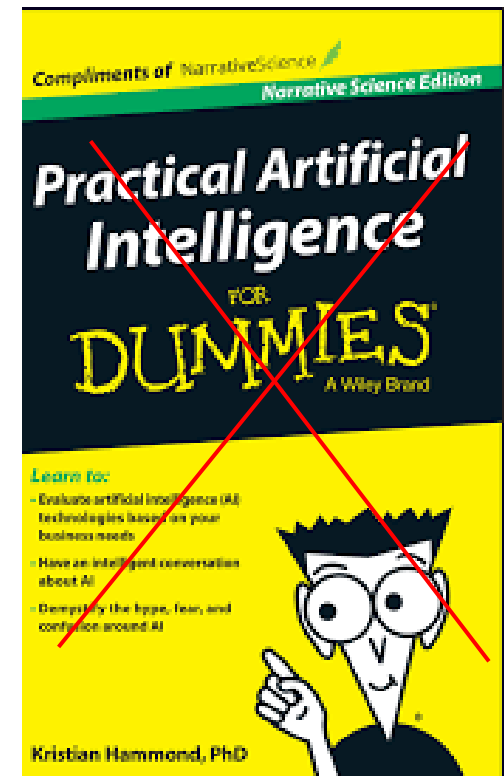


Toolbox #4: Patience and Theory

Since this is not just a «how-to-create-a-neural-network course» or «step-by-step guide to...» we need to face some relevant theory topics before to start the lab/coding activities.



Fabio Scotti - Università degli Studi di Milano -
Informatics for industrial applications and robotics



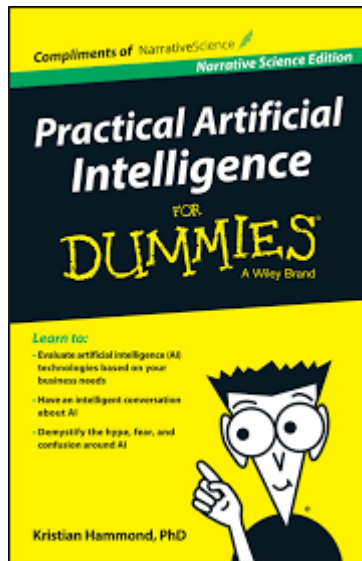


How to study this course?

From superficiality to interiorization

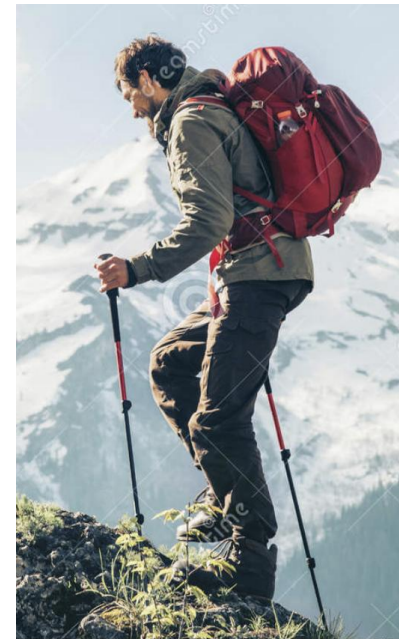
How to learn the AI theory?

Working and designing with
“the book” in your hands?



The book really exists,
but please do not buy it...

... or to be hands-free and the
knowledge is your backpack?



Real knowledge requires **interiorization!**

How can I get interiorization?



Especially in the machine learning field you must

1. Study the **theory** and the related **working hypothesis**
2. Study as **many use-cases** as possible to extend your point of view
3. Apply the **theory on real applications**
4. Make a **fair discussion and comparison** every time is possible.
 - Ask you “Why this and not that?”
 - Example: “What is the model really learning?”

How can I get interiorization? (cont.)

- Create your “portfolio” of **models** and **learning methods** (as large as possible)
- **Keep you informed** on the *State Of the Art* (SOA) and the trends
- Always **choose** from different solutions



How can I get interiorization? (cont.)

- Do not consider only accuracy
 - There are a wide range of **figures of merit** to consider in your design (in the next lessons)
- Improving your project by **building on previous experience and discoveries**
 - Standing on the shoulders of giants!
The analysis of the SOA is fundamental.



Main points



- Prepare your toolboxes!
(Matlab, Colab, and the other resources)
- In this field, knowledge **interiorization** is needed to create **skills**
- Taxonomy of machine learning methods
- Focus on
 - Narrow AI
 - Regression
 - Classification
 - Clustering
 - Deductive and inductive learning
- Overview of AI applied to sensors
 - Intelligent sensors
 - Networks of sensors
 - Advanced instrumentation and **virtual sensors**